

Evaluating VIXY Option Strategies: A Backtesting and Implied Borrow Cost Framework

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1 Introduction

The VIXY (ProShares VIX Short-Term Futures ETF) is a fund that tracks market volatility. It is designed to surge in value during periods of sudden macroeconomic stress or market uncertainty. To maintain a constant one-month maturity, the fund must roll its futures contracts on a daily basis. Since VIXY is heavily impacted by contango (market state where longer-dated VIX futures are more expensive than near-term ones), VIXY must sell cheaper, expiring futures, and buy more expensive next-month contracts. As a result, this negative roll yield (selling low and buying high) causes the fund's net asset value to consistently decay over time.

Due to VIXY's structural downward drift, VIXY has become a big target for short sellers. However, high demand to short the ETN leads to limited availability. Brokers classify VIXY as hard to borrow and charge high daily borrow fees that often neutralize profitability of the trade. To bypass the brutal broker fees, institutional traders go to the options market.

Our project backtests a three year trading strategy that utilizes synthetic shorting. To create a synthetic short, we pair two options together that mimic a normal short. By buying a put and selling a call at the same strike price, we build a trade that makes money when VIXY drops and loses money when it climbs. This allows for the same results as a traditional short, without the struggle of finding a broker that has shares to lend.

On another note, the options market is highly efficient. Since many people want to buy "insurance" against a market crash, the price of VIXY options gets pushed higher. This creates a hidden fee inside the price of the options. By looking at three years of historical data, our project uses the Put-Call Parity to figure out exactly how much the borrow cost is. Our goal is to see if VIXY's natural decay is big enough to actually make a profit, or if the borrow cost is so high that it makes the trade a waste of time.

2 Sources and Data

This analysis uses option market data and underlying price data for VIXY. The options dataset is sourced from OptionMetrics via the Wharton Research Data Services (WRDS) platform. OptionMetrics provides cleaned U.S. equity options data, including bid and ask quotes, strike prices, expiration dates, and option identifiers. The historical price for VIXY is obtained from Nasdaq. The dataset provides daily closing prices of VIXY, which are used as the spot price in the analysis. [Opt26], [Nas26]

In addition, Bloomberg terminal snapshots are used for cross-checking market-implied pricing relationships at a fixed date: a Treasury yield chart, and matched put and call option screens for VIXY.

3 Comparison of VIX and VIXY

The reason VIXY tends to go down over time while the VIX does not lies in what each of them actually represents. The VIX itself is not an asset you can buy or hold. It is a calculated index that reflects the market's expectation of future volatility based on S&P 500 options. Because it is just a measure, it moves up and down with market conditions, often spiking during stress and then falling back, but it does not have any built-in mechanism that causes it to decay over time.

By contrast, VIXY is an exchange-traded fund that holds VIX futures contracts rather than the VIX index itself. These futures have expiration dates, so the fund cannot simply hold them indefinitely. Instead, it must continuously "roll" its position, selling contracts that are about to expire and buying contracts with a later maturity. This rolling process is the key to understanding the long-term behavior of VIXY.

Most of the time, the VIX futures curve is in a state known as contango, where longer-dated futures are priced higher than near-term ones. When VIXY rolls its position in this environment, it systematically sells cheaper contracts and buys more expensive ones. This creates a persistent loss, often described as roll decay. Even if the level of volatility does not change much, this repeated "buy high, sell low" process causes the value of the fund to decline over time.

This effect accumulates gradually and gives VIXY a strong downward drift over the long run. It is not that volatility is disappearing, but rather that maintaining exposure to volatility through futures is costly. In a sense, holding VIXY is like continuously paying an insurance premium. That cost is embedded in the futures prices and is realized through the rolling process.

The VIX index, on the other hand, does not experience this decay because it does not involve holding or rolling any contracts. It is simply a snapshot of implied volatility at a given moment. As a result, it can remain stable or fluctuate around a long-term average without trending downward in the same way.

This structural difference is also why many traders are interested in shorting VIXY. Because the product naturally loses value over time, there is strong demand to bet against it. That demand contributes to the borrowing costs you are estimating in your analysis. In this way, the persistent decline of VIXY is directly tied to both its construction and the economic forces reflected in the options market.

ProShares VIX Short-Term Futures ETF (VIXY) ☆

28.39 +0.46 (+1.65%) 28.30 -0.09 (-0.32%)

At close: 4:00:00 PM EDT

After hours: 5:05:01 PM EDT

Trade VIXY on Coinbase

Trading disclosure ⓘ



Figure 1: Logarithmic Path of VIXY (April 2016 – April 2026), demonstrating significant long-term value decay. [Yah26]

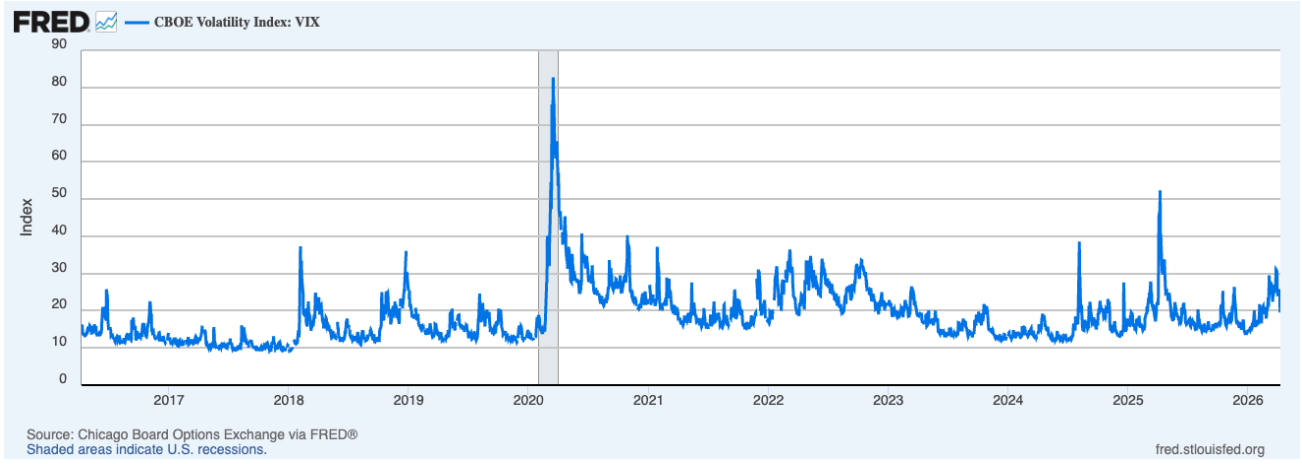


Figure 2: VIX Index (April 2016 - April 2026), demonstrating stable and fluctuations around a long-term average [Chi26]

4 Strategies Backtesting

4.1 Synthetic Short

The synthetic short strategy is designed to replicate a short position in VIXY by combining options rather than directly shorting the ETF. This is implemented by buying one put and selling one call with the same expiration and, as closely as possible, the same strike, typically chosen near at-the-money using delta. The intuition is that this position should behave similarly to shorting VIXY because it benefits when VIXY declines and loses when VIXY rises. In the backtest, the strategy is rolled as expiration approaches so that the position maintains exposure to short-term VIXY movements. This makes it the closest options-based approximation to a direct short, although in practice its performance can differ because of bid-ask spreads, imperfect strike matching, missing quotes, and the fact that options also embed volatility and time-decay effects.

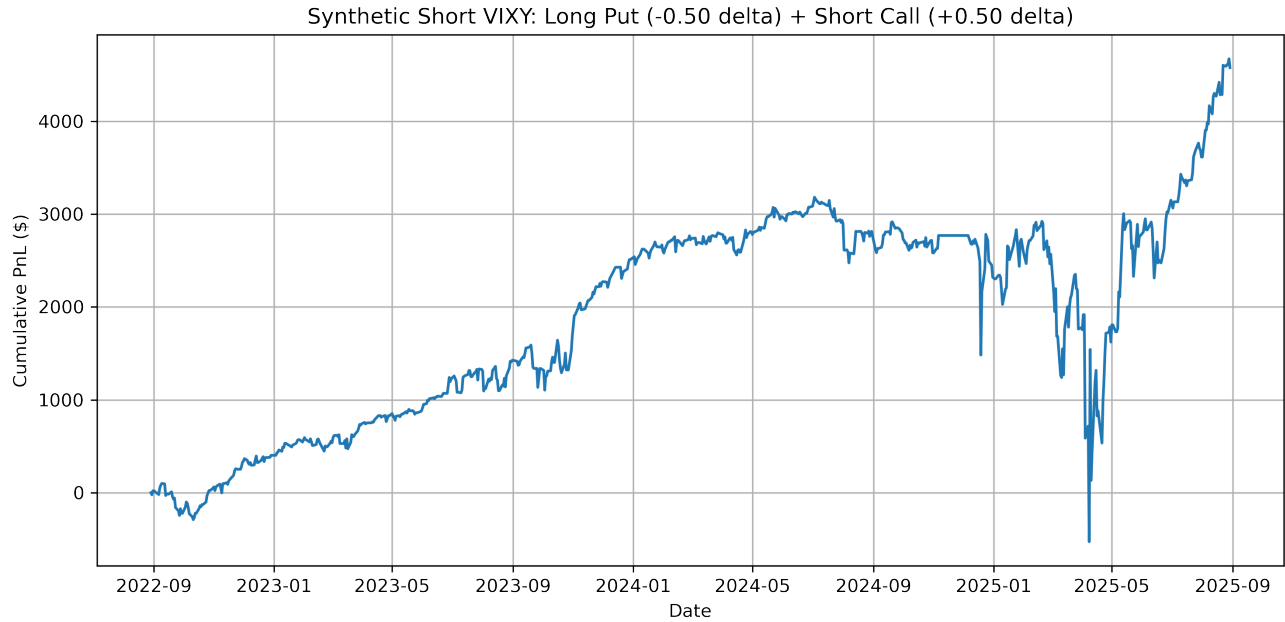


Figure 3: Cumulative Profit and Loss (PnL) for a Synthetic Short VIXY options strategy (Long -0.50 delta Put, Short +0.50 delta Call) from September 2022 through September 2025. The strategy exhibits steady, upward returns through late 2024, followed by a severe drawdown (tariffs) event in 2025 before a sharp recovery.

4.2 Short Call

The short call strategy is simpler and more directional. It is implemented by selling a call option on VIXY, usually one with delta close to +0.50 and expiration near 30 days. This strategy profits when VIXY stays flat or declines, because the call premium decays over time and the option is less likely to finish in the money. Compared with the synthetic short, the

short call usually generates a similar PnL pattern because both are effectively bearish on VIXY, but the short call tends to produce lower overall returns and carries significant upside risk if VIXY spikes sharply. Its main source of return is option premium collection rather than direct replication of the ETF's price decline, so it often underperforms a pure short-VIXY position during long periods of steady VIXY decay.



Figure 4: Cumulative Profit and Loss (PnL) for a Short Call VIXY options strategy (Short +0.50 delta Call) from September 2022 through September 2025. The strategy exhibits a PnL trajectory similar to Synthetic Short except for a consistent lower return.

4.3 Iron Condor

The iron condor strategy is the most risk-controlled of the three. It is built by combining a short call spread and a short put spread on the same expiration: selling an out-of-the-money call and put while simultaneously buying a further out-of-the-money call and put as protection. This creates a position that profits most when VIXY remains within a range and all four options expire with limited intrinsic value. Because both the upside and downside are hedged, the iron condor has capped losses as well as capped gains, which makes its cumulative PnL path much smoother than the synthetic short or short call strategies. However, this protection comes at the cost of lower total return, since the strategy sacrifices some premium income in exchange for reduced tail risk. In the backtest, the iron condor therefore appears as the most stable but also the least aggressive way to express a short-volatility view on VIXY.

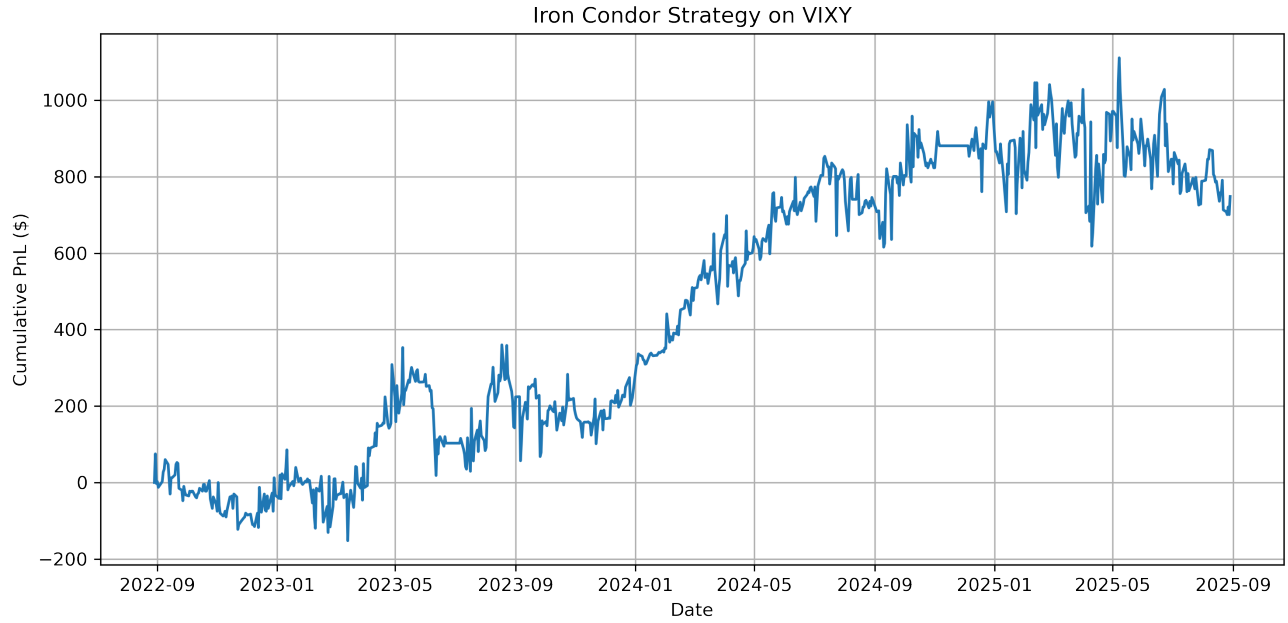


Figure 5: Cumulative profit and loss (PnL) of a short put short call VIXY options strategy from September 2022 to September 2025. The strategy exhibits a smoother PnL trajectory compared to both the synthetic short and short-call strategies, reflecting its limited risk and return profile, but results in lower overall returns.

4.4 Short Straddle

Although the short put and short call (short straddle) strategy appears relatively stable in the graph, this stability can be somewhat misleading. The strategy benefits from consistently collecting option premium as both options decay over time, especially during periods when VIXY remains within a narrow range, which creates a smooth and gradually increasing PnL trajectory. However, this apparent stability masks the strategy's inherent risk, as it is exposed to large losses in both directions if VIXY experiences a significant move. In addition, the backtest may understate extreme losses due to factors such as missing quotes, mid-price assumptions, and forced exits when data becomes unavailable, all of which can smooth out sharp drawdowns. As a result, while the short straddle appears less volatile than other strategies in the sample period, it remains a high-risk, short-volatility strategy with potentially severe tail risk during periods of market stress.

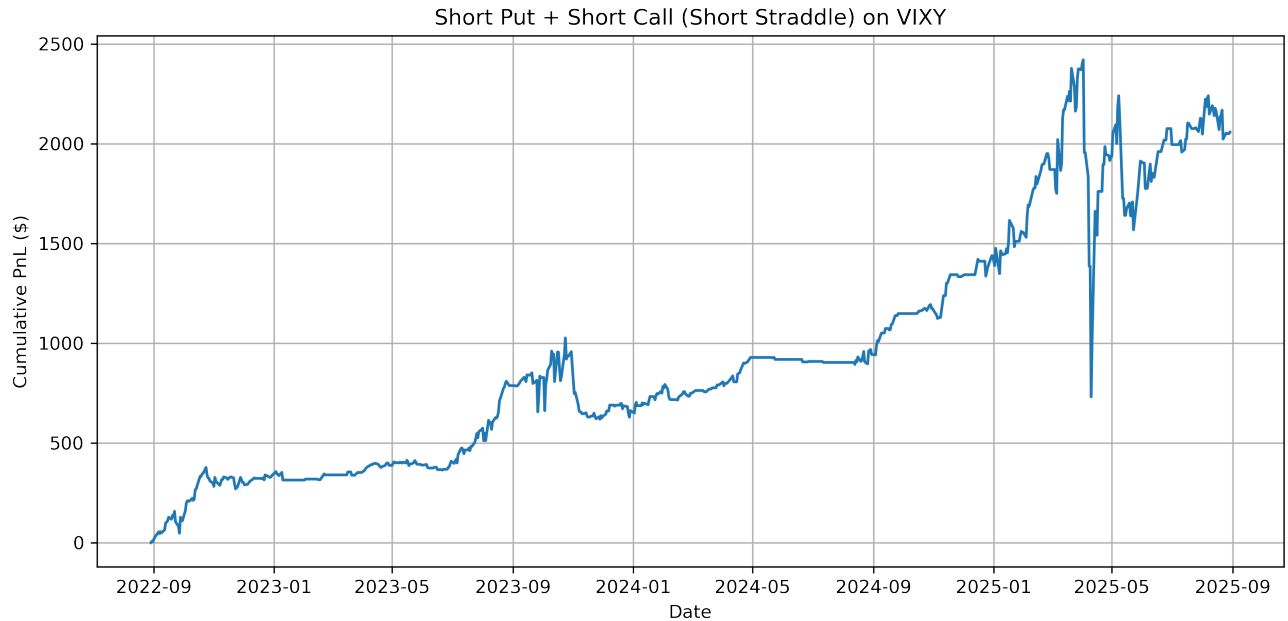


Figure 6: Cumulative profit and loss (PnL) of a short put and short call (short straddle) VIXY options strategy from September 2022 to September 2025. The strategy follows a similar PnL pattern to the synthetic short and short-call strategies, but shows less return and less volatility.

4.5 Comparison and Analysis

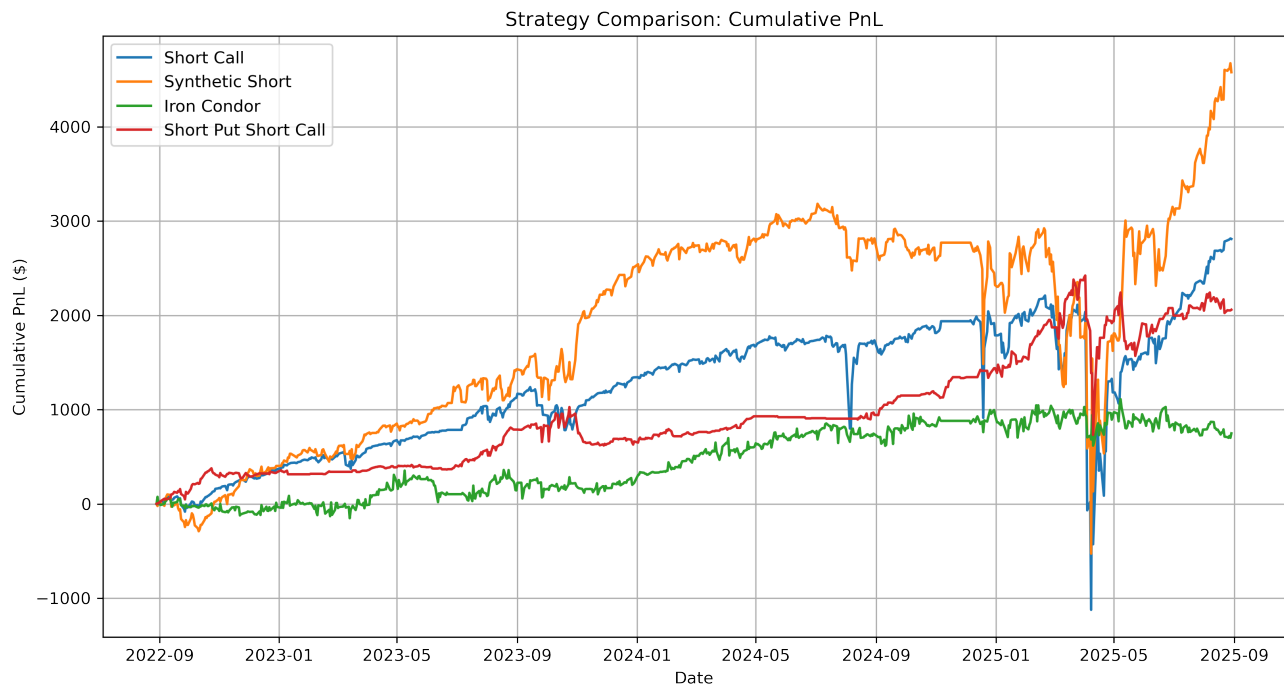


Figure 7: Strategy Comparison: Cumulative PnL

Consolidated Risk and Return Metrics Across VIXY Options Strategies (Aug 2022 - Aug 2025)

Metric	Synthetic Short	Short Call	Iron Condor	Short Put + Short Call
Total PnL (\$)	4,577.50	2,809.50	748.00	2,059.00
Average Trade PnL (\$)	59.17	37.53	11.80	13.32
Win Rate (%)	68.18%	76.12%	56.72%	37.66%
Trade PnL Std. Dev. (\$)	256.24	191.48	59.40	85.03
Annualized Sharpe Ratio				
2022-08-29 to 2023-08-29	1.96	2.63	0.31	0.31
2023-08-29 to 2024-08-29	1.47	0.55	0.66	0.79
2024-08-29 to 2025-08-29	0.46	0.29	-0.02	0.21

Table 1 shows the performance and risk metrics for four distinct option strategies applied to the VIXY over a three-year period from August 2022 to August 2025. All data presented reflects the performance of a position size of exactly one contract.

Key Findings:

1. Relationship Between Risk and Reward (Per Contract)

The data shows a direct relationship between absolute returns and volatility. On a per-contract basis, the Synthetic Short generated the highest profit (\$4,577.50) but required absorbing the highest per-trade volatility (standard deviation of \$256.24). On the other hand, the Iron Condor strategy had decreased per-contract volatility with a standard deviation of only \$59.40, but at the cost of overall returns, generating only \$748.00 over the three year period.

2. Win Rate vs. Profitability per Unit

The Short Call strategy shows the highest win rate of the strategies at 76.12%, closing out more than three-quarters of its individual contracts for a profit. However, this strategy was outperformed in Total PnL by the Synthetic Short, which has a win rate nearly ten percentage points lower (68.18%). This shows that while the Short Call wins more frequently, the Synthetic Short captures larger profit margins per winning contract, averaging \$59.17 per contract compared to \$37.53 for the Short Call.

3. Declining Strategy Efficiency Over Time

The Annualized Sharpe Ratios show that trading individual contracts became significantly less effective over the three-year period. In the first year (2022-2023), the volatility of shorts was highly profitable relative to the risk taken, with the Short Call (2.63) and the Synthetic Short (1.96) showing strong results.

However, by the third year (2024-2025), the performance of each contract dropped sharply across all four strategies. The Short Call's efficiency dropped to 0.29, and the Iron Condor saw its risk-adjusted returns turn negative (-0.02). This downward trend suggests that market conditions changed, making it a lot harder to earn a profit per contract without facing higher levels of risk.

4. Straddle Anomaly: High Profit with a Low Win Rate

The Short Put + Short Call (Straddle) stands out because it made a solid total profit (\$2,059.00) although it has the lowest win rate (37.66%). This shows that while the strategy loses most of the time, its winning trades are big enough to make up for the losses. Additionally, this is the only strategy where the risk-adjusted returns (Sharpe Ratio) improved in the second year (0.79) instead of decreasing immediately.

5 Implied Borrow Cost of VIXY

The borrow cost of an asset reflects the cost of shorting that asset in the securities lending market. For exchange-traded products such as VIXY (ProShares VIX Short-Term Futures ETF), borrow costs can be substantial due to persistent demand for short exposure. [Duf92]

In practice, borrow costs are not always directly observable. However, they can be inferred from option prices using put-call parity. This approach has been widely used in the literature to extract implied financing rates and shorting costs from options markets. [GPP09], [BW04]

For VIX-linked ETFs, understanding borrow cost is particularly important because these products exhibit strong negative carry due to futures roll decay. As a result, investors frequently seek short exposure, potentially driving up borrowing costs.

The standard single-maturity approach starts from the no-arbitrage relation

$$C - P = (F - K)e^{-rT}, \quad (1)$$

where C and P denote the call and put prices with the same strike K and maturity T , r is the risk-free rate, and F is the forward price. Rearranging gives

$$b = -\frac{1}{T} \log\left(\frac{C - P + Ke^{-rT}}{S}\right). \quad (2)$$

Although (2) is the usual formula, it depends directly on the observed spot price S . In our setting, this can introduce unnecessary mismatch. The option prices and the reported spot price may not be perfectly synchronized in time; closing option quotes and closing ETF prices can reflect slightly different market states; and using a separate spot data source may add measurement noise or timing inconsistencies. As a result, the inferred borrow cost may partly reflect spot-price mismatch rather than the information contained in the options themselves.

For this reason, we use a two-horizon forward-based formula that eliminates the spot price altogether. Instead of comparing one option-implied forward to the observed spot, we compare two option-implied discounted forwards at different maturities. This keeps the estimation entirely within the options market and reduces noise coming from spot-option misalignment.

5.1 Derivation of the Two-Horizon Formula

Let (C_1, P_1, K_1, T_1) and (C_2, P_2, K_2, T_2) denote matched call-put pairs observed on the same trading date, but with two different maturities $T_1 < T_2$. For each maturity, put-call parity implies

$$C_i - P_i = (F_i - K_i)e^{-rT_i}, \quad i = 1, 2. \quad (3)$$

where:

- C_i : mid-price of the call option at maturity T_i ,

- P_i : mid-price of the put option at maturity T_i ,
- K_i : strike price used for the matched call-put pair at maturity T_i ,
- T_i : time to maturity in years, for $i = 1, 2$,
- r : continuously compounded risk-free interest rate,
- F_i : forward price implied by put-call parity at maturity T_i ,

Rearranging,

$$C_i - P_i + K_i e^{-rT_i} = F_i e^{-rT_i}. \quad (4)$$

Now assume the forward price at maturity T_i is generated by the same spot price S and annualized borrow cost b :

$$F_i = S e^{(r-b)T_i}, \quad i = 1, 2. \quad (5)$$

Substituting into (4) gives

$$C_i - P_i + K_i e^{-rT_i} = S e^{(r-b)T_i} e^{-rT_i} = S e^{-bT_i}. \quad (6)$$

Applying (6) at the two maturities:

$$C_1 - P_1 + K_1 e^{-rT_1} = S e^{-bT_1}, \quad (7)$$

$$C_2 - P_2 + K_2 e^{-rT_2} = S e^{-bT_2}. \quad (8)$$

Taking the ratio of (7) and (8) eliminates the unobserved spot price S :

$$\frac{C_1 - P_1 + K_1 e^{-rT_1}}{C_2 - P_2 + K_2 e^{-rT_2}} = \frac{S e^{-bT_1}}{S e^{-bT_2}} = e^{b(T_2 - T_1)}. \quad (9)$$

Taking logarithms yields

$$\log(C_1 - P_1 + K_1 e^{-rT_1}) - \log(C_2 - P_2 + K_2 e^{-rT_2}) = b(T_2 - T_1), \quad (10)$$

so the implied borrow cost is

$$b = \frac{\log(C_1 - P_1 + K_1 e^{-rT_1}) - \log(C_2 - P_2 + K_2 e^{-rT_2})}{T_2 - T_1}. \quad (11)$$

5.2 Interpretation

Equation (11) is the formula used in our time series analysis. It measures the borrow cost from the relative behavior of two option-implied discounted forwards across maturities, rather than from a single maturity relative to the observed spot price. The main advantage is that the spot price cancels out, so the estimate is less sensitive to stale spot data, timing differences between the ETF and option quotes, and other data-alignment issues.

Intuitively, if borrow costs are positive, then the discounted forward quantity

$$C - P + K e^{-rT} = S e^{-bT}$$

declines with maturity T , since a larger borrow cost depresses the forward value more strongly at longer horizons. Comparing a shorter-horizon and a longer-horizon option-implied forward therefore reveals the rate at which borrow costs accumulate over time.

5.3 Single-Option Borrow Cost Implementation

Before constructing the two-horizon estimator, we verify that the standard single-maturity formula (2) produces sensible results using Bloomberg market data from a single trading date.

The Treasury yield plot provides the funding anchor for discounting and forward-implied calculations. Since option-implied borrow estimates are sensitive to the interest-rate input, we used a same-day Treasury reference. Comparing the Bloomberg put and call screens at the same strike and maturity gives a direct parity-based consistency check: the relative pricing between puts and calls reveals the market-implied cost of carrying a short position in VIXY. In our setting, the option price difference implies a non-trivial borrow component, consistent with the persistent short demand discussed earlier.

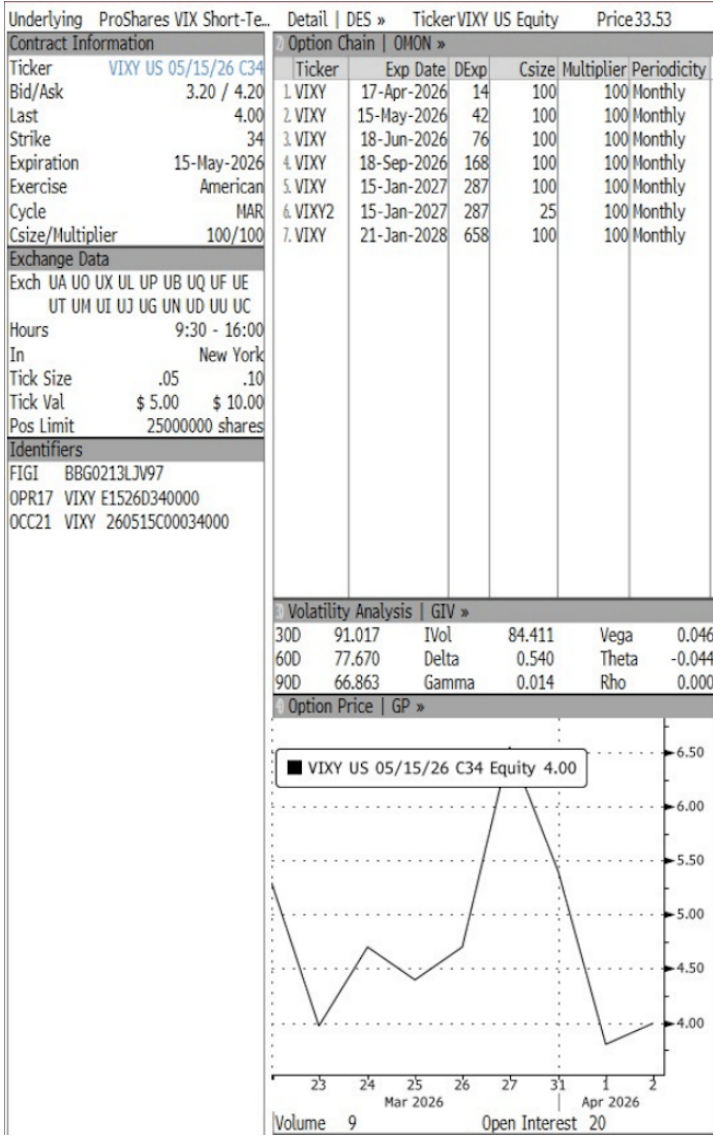


Figure 8: Bloomberg VIXY call option screen at the 34 strike, maturing May 15, 2026.

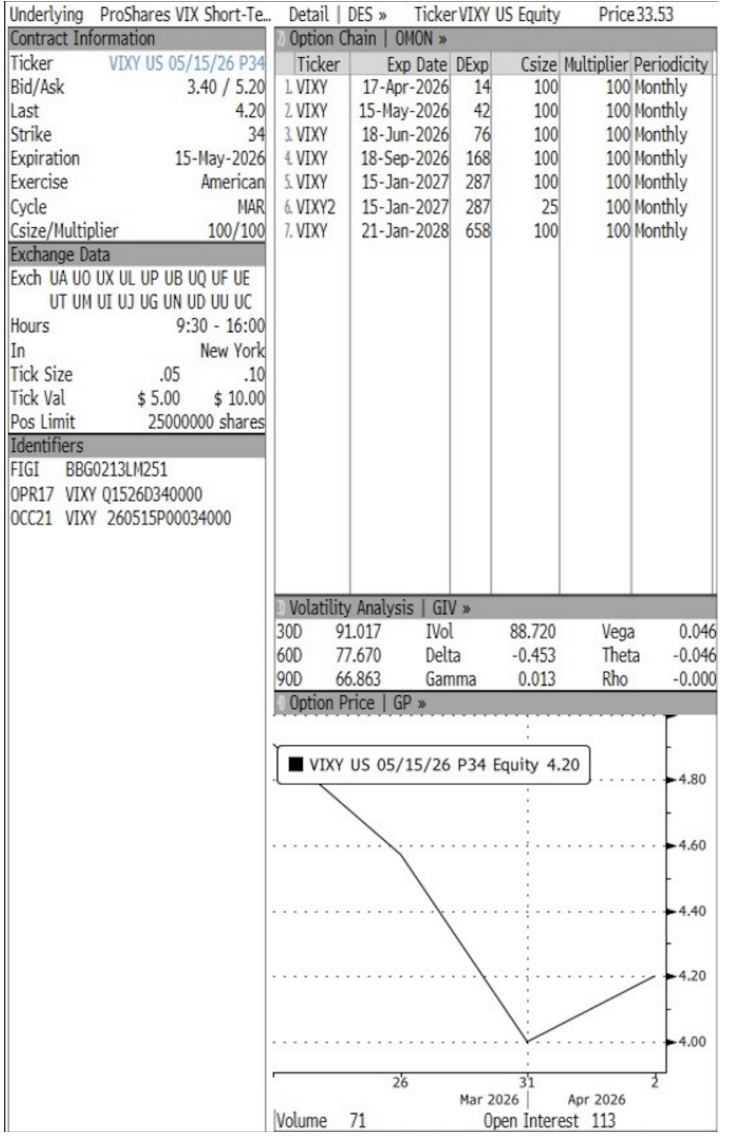


Figure 9: Bloomberg VIXY put option screen at the 34 strike, maturing May 15, 2026.

Using the Bloomberg single-date inputs (spot price $S = 33.53$, strike $K = 34$, call mid $C = 3.7$, put mid $P = 4.3$, days to expiry $T = 45/365$, and risk-free rate $r = 0.03683$), we apply (2) to compute the implied borrow cost. We first obtain the forward price $F = K + (C - P)e^{rT}$ from put-call parity and then solve for b . The result is an implied forward price of **\$33.40** and an implied borrow cost of **6.90%**. The positive borrow estimate is consistent with persistent demand to short VIXY and confirms that the single-maturity approach in (2) captures a meaningful borrow component. However, as noted earlier, this estimate depends on the observed spot price S , motivating the two-horizon formula developed in Section 5.1.

5.4 Two-Horizon Empirical Implementation

In the empirical analysis, we construct matched call-put pairs from OptionMetrics data using options with the same trading date, strike, and expiration. Mid-prices are computed as the average of bid and ask quotes, and observations with wide bid-ask spreads are excluded to reduce noise.

We fix the short horizon at

$$T_1 = 1 \text{ month},$$

and consider three longer horizons:

$$T_2 \in \{6 \text{ months}, 12 \text{ months}\}.$$

For each choice of T_2 , we compute cross-sectional implied borrow costs across strikes and compare the stability of the estimates. Shorter horizons exhibit higher variability and are more sensitive to microstructure noise, while longer horizons (e.g., 12 months) provide smoother estimates but may incorporate additional term structure effects.

Based on this comparison, we select the 6-month horizon as a balance between stability and economic relevance. The 6-month maturity provides sufficiently smooth and consistent estimates while remaining responsive to medium-term borrowing conditions. All reported implied borrow cost results are therefore based on

$$(T_1, T_2) = (1 \text{ month}, 6 \text{ months}).$$

5.5 Term Structure and Time Series of Borrow Cost

The time series of borrow cost describes how implied borrowing rates vary across maturities. For each date, we compute the median borrow cost for each expiration and plot it as a function of time to maturity.

Empirically, we find that the term structure of VIXY borrow cost is relatively stable across time, with values remaining persistently elevated. This suggests that the demand for short exposure is not limited to a specific horizon but is instead a structural feature of the market.

A stable or slightly upward-sloping term structure indicates that investors are consistently willing to pay a premium to short VIXY over both short and long-term horizons.

Our results show that the implied borrowing cost of VIXY is persistently high during the sample period. This is consistent with the economic intuition that VIXY tends to decline over time due to the roll cost associated with VIX futures in contango. As a result:

- Investors are incentivized to short VIXY,
- Short demand drives up borrowing costs,
- This is reflected in option prices via put–call parity.

The relatively stable term structure further suggests that this effect is systematic rather than episodic. In other words, the cost of shorting VIXY is consistently priced into the options market.

For VIXY, the persistently high and stable borrow cost reflects structural features of volatility-linked products, including negative carry and strong investor demand for hedging or speculative short positions.

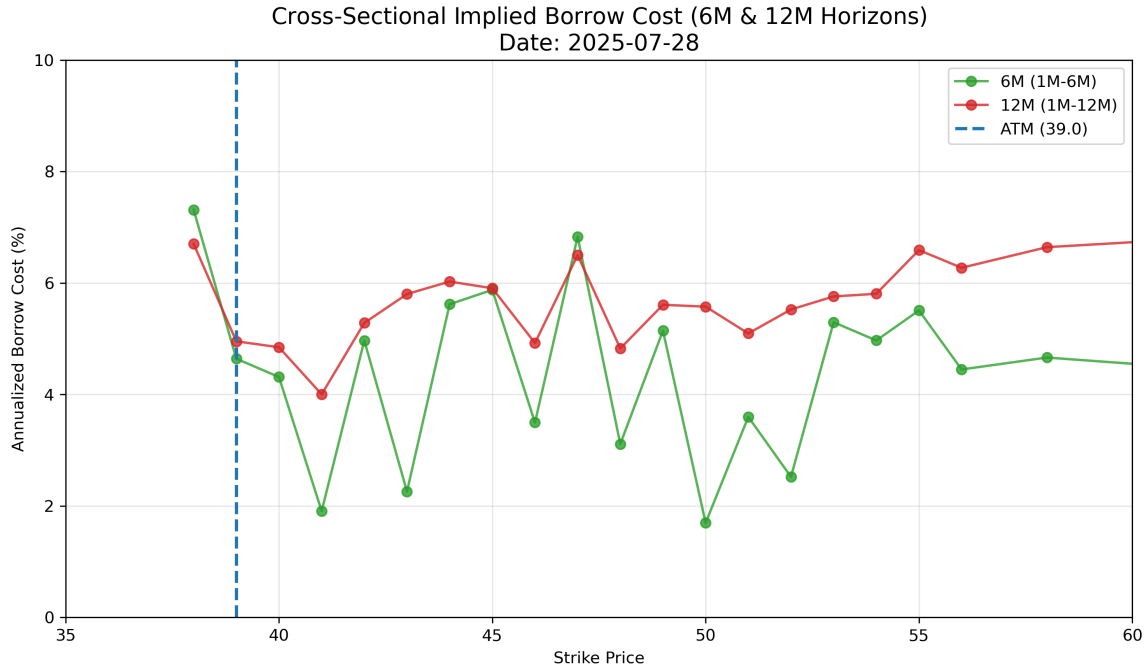


Figure 10: Cross-sectional implied borrow cost for VIXY across strike prices for 6-month and 12-month horizons.

The term structure effect is evident in the cross-section: shorter maturities display higher variability and occasional outliers, whereas longer maturities yield smoother and more consistent estimates. In addition, the 12-month implied borrow cost lies consistently above the 6-month estimates across most strikes, indicating an upward-sloping term structure. This suggests that investors are willing to pay a higher premium to short VIXY at longer horizons.

At the same time, longer-dated options provide a more reliable measure of the underlying borrow cost by averaging out short-term distortions. However, very long maturities, such as 12 months, may incorporate additional term premia, liquidity effects, and model uncertainty that can bias the estimate away from the true borrow cost. As a result, the 6-month horizon

provides a balance between stability and relevance: it is sufficiently long to reduce noise and microstructure effects, yet short enough to remain closely tied to the economically meaningful borrow cost faced by market participants. Therefore, we use the 6-month implied borrow cost as our primary measure in subsequent analysis.

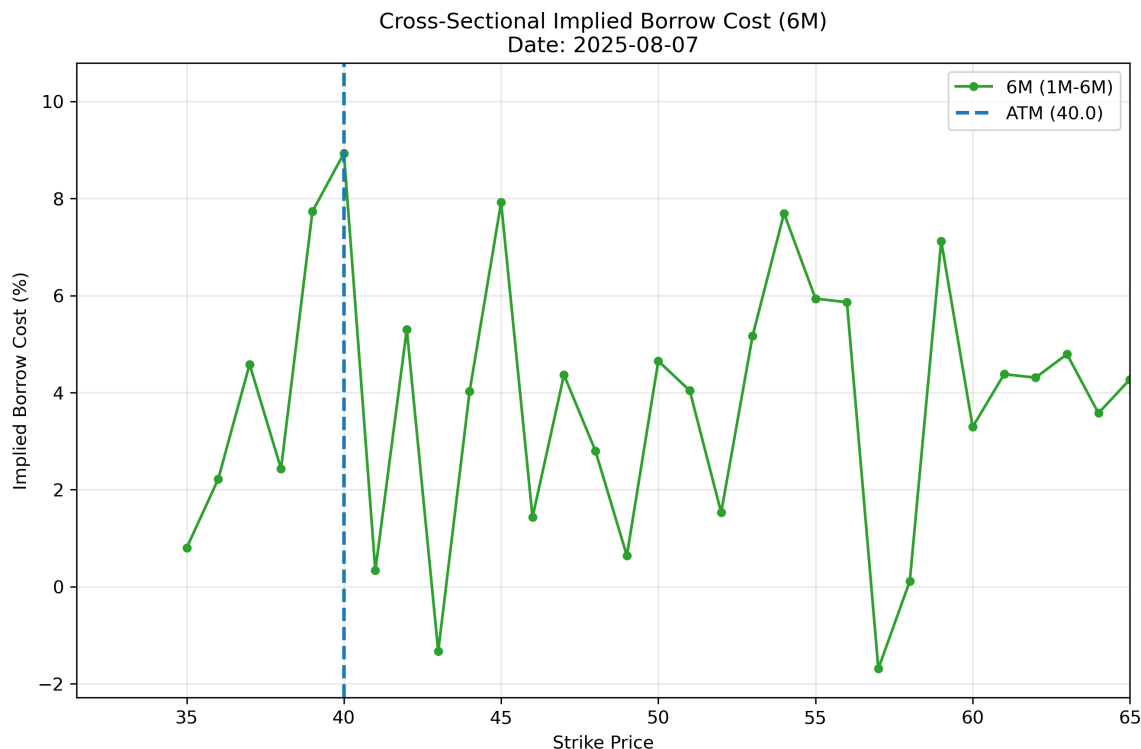


Figure 11: Cross-sectional implied borrow cost for VIXY across strike prices using a 6-month horizon on August 7, 2025. The dashed vertical line denotes the at-the-money (ATM) level. Compared to shorter maturities, the estimates are smoother and exhibit reduced noise, indicating greater stability in longer-dated option-implied borrow cost measures.

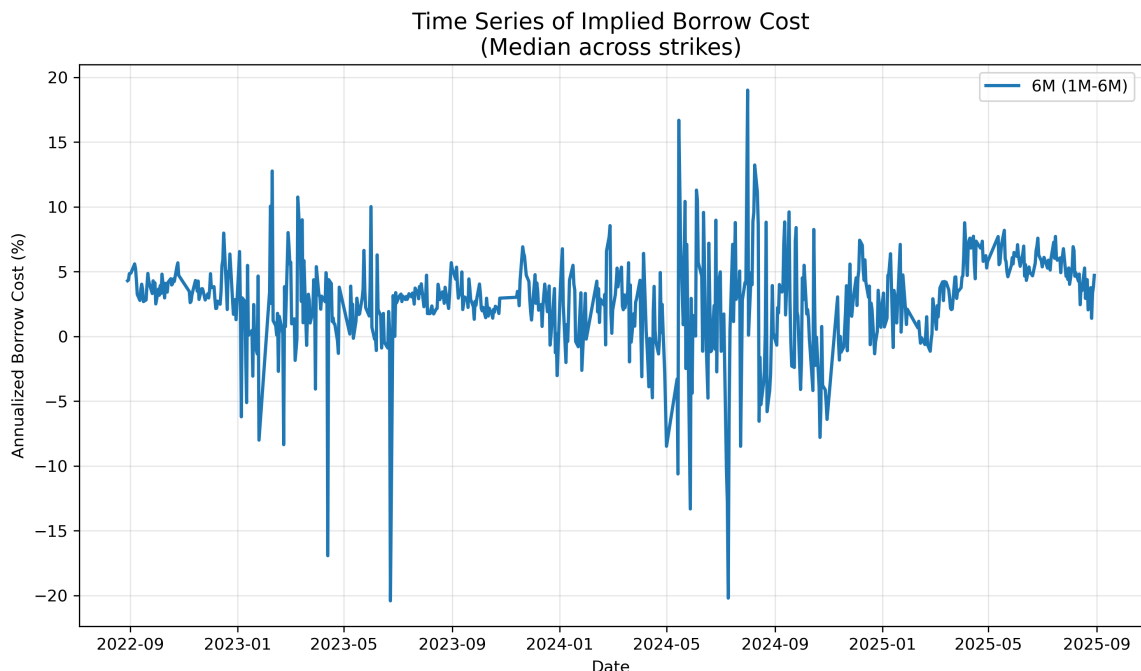


Figure 12: Time Series of implied borrow cost for VIXY based on a 6-month horizon, computed as the median across strikes. The borrow cost displays a generally stable, mildly upward-sloping pattern over time, suggesting persistent demand for short exposure, with occasional spikes reflecting transient market stress.

Figure 11 compares the cross-sectional implied borrow cost across strike prices for 6-month and 12-month horizons, revealing a clear maturity effect. The 12-month estimates are smoother and more stable across strikes, while the 6-month estimates exhibit greater variability and occasional outliers. At the same time, the two curves track each other closely in shape across strikes, suggesting that the estimates reflect a common underlying economic signal rather than independent noise. Notably, the longer-horizon estimates lie consistently above the shorter-horizon ones, indicating an upward-sloping implied borrow cost.

This pattern reflects a trade-off between statistical noise and economic relevance. Longer maturities incorporate more information and average out short-term fluctuations, leading to smoother estimates, but may also embed additional term premia, liquidity effects, and model uncertainty that can bias the inferred borrow cost away from its true economic value [Duf92; GPP09]. In contrast, shorter maturities are more responsive to current market conditions but are more sensitive to microstructure noise.

Across strikes at a fixed maturity, the implied borrow cost exhibits a well-defined structure. Estimates are most stable around the at-the-money (ATM) region, where option liquidity is highest and price signals are most reliable, while dispersion increases for deep in-the-money and out-of-the-money options. This behavior is consistent with well-documented option market microstructure effects, as less liquid strikes tend to have wider bid-ask spreads and noisier pricing [BCC00]. Despite this variation, the central tendency of the cross-section remains positive across strikes.

Extending beyond the cross-section, the time series of the median implied borrow cost further reinforces this interpretation. The borrow cost remains persistently positive over the sample period, typically centered around a few percentage points, with occasional spikes and deviations associated with volatility stress events or temporary liquidity distortions. If short demand for VIXY were purely episodic, the distribution would be centered near zero outside of crisis periods. Instead, the consistently positive level indicates that shorting VIXY is systematically costly.

Taken together, these results show that the implied borrow cost is economically consistent and robust, reflecting a structural equilibrium in the market driven by sustained demand for short volatility exposure and the negative carry embedded in VIX futures.

6 Conclusion

VIXY’s natural decay is large enough to generate profit; however, high implied borrow costs and exposure to severe drawdowns during volatility spikes significantly reduce the strategy’s efficiency over time, as reflected in a declining Sharpe ratio. Our backtesting results confirm that trading on VIXY’s downward drift is economically viable, as all four options strategies considered—Synthetic Short, Short Call, Iron Condor, and Short Straddle—produced positive total profits over the three-year sample period. Among these, the Synthetic Short strategy was the most profitable, generating \$4,577.50 per contract.

Despite this profitability, the options market appears highly efficient in pricing shorting demand directly into option premiums. Using put-call parity, we find that the implied borrow cost for VIXY remains consistently elevated throughout the sample period, with a relatively stable time series. This suggests that the high cost of shorting VIXY is not temporary but rather a structural feature of the market, driven by persistent investor demand to short volatility-linked products. As a result, a hidden cost is continuously embedded in option prices, effectively reducing the net profitability of these strategies.

Moreover, the profitability of these trades comes with substantial risk and evolving market conditions. There is a clear relationship between returns and volatility exposure: strategies with higher absolute returns, such as the Synthetic Short, also exhibit greater per-trade volatility. Over time, this risk becomes increasingly significant, as evidenced by the declining Sharpe ratios across all strategies. While the first year (2022–2023) delivered strong risk-adjusted performance, this deteriorated sharply by the third year (2024–2025). A key contributor to this decline was the 2025 tariff event, which triggered a spike in volatility and a corresponding surge in VIXY, leading to significant drawdowns and a marked reduction in overall risk-adjusted returns.

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A Synthetic Short VIXY Trade Log

– PERFORMANCE SUMMARY –

Note: All metrics below are calculated based on a position size of exactly one contract.

Total PnL (\$): 4577.50

Number of completed trades: 66

Average trade PnL (\$): 59.17

Win rate: 68.18%

Trade PnL Standard Deviation (\$): 256.24

Table 1: Synthetic Short VIXY Trade Log (Aug 2022 - Aug 2025)

ID	Entry	Exit	Expiry	DTE (In)	DTE (Out)	Call ID	Put ID	Call K	Put K	Call Δ	Put Δ	C(In)	P(In)	C(Out)	P(Out)	PnL (\$)
1	2022-08-29	2022-09-15	2022-09-30	32	15	148915193	148915214	15.0	15.0	0.498	-0.506	1.225	1.550	0.750	0.925	-15.00
2	2022-09-16	2022-09-29	2022-10-14	28	15	149412453	149412474	15.0	15.0	0.497	-0.491	0.975	1.375	2.050	0.375	-207.50
3	2022-09-30	2022-10-13	2022-10-28	28	15	149502993	149503014	17.5	17.5	0.493	-0.497	1.350	1.925	0.750	1.325	0.00
4	2022-10-14	2022-10-27	2022-11-11	28	15	150018471	150018492	18.0	18.0	0.485	-0.519	1.325	1.900	0.175	3.200	245.00
5	2022-10-28	2022-11-10	2022-11-25	28	15	150107797	150107818	14.5	14.5	0.529	-0.470	0.800	0.750	0.325	1.025	75.00
6	2022-11-11	2022-11-25	2022-12-09	28	14	150535893	150535914	14.0	14.0	0.489	-0.519	0.725	0.875	0.200	1.900	155.00
7	2022-11-28	2022-12-15	2022-12-30	32	15	150746249	150746270	13.0	13.0	0.497	-0.511	0.850	1.025	0.250	1.150	72.50
8	2022-12-16	2022-12-29	2023-01-13	28	15	151179420	151179441	12.5	12.5	0.460	-0.535	0.650	1.025	0.250	1.400	77.50
9	2022-12-30	2023-01-12	2023-01-27	28	15	151278809	151278830	11.5	11.5	0.498	-0.498	0.650	0.800	0.180	1.625	129.50
10	2023-01-13	2023-01-26	2023-02-10	28	15	151754021	151754042	10.0	10.0	0.481	-0.520	0.605	0.860	0.310	0.955	39.00
11	2023-01-27	2023-02-09	2023-02-24	28	15	151837575	151837596	9.0	9.0	0.524	-0.492	0.605	0.510	0.905	0.190	-62.00
12	2023-02-10	2023-02-23	2023-03-10	28	15	152303844	152303861	10.0	10.0	0.481	-0.506	0.565	0.860	0.460	0.600	-15.50
13	2023-02-24	2023-03-09	2023-03-24	28	15	152406435	152406452	10.0	10.0	0.522	-0.477	0.830	0.925	0.405	0.860	36.00
14	2023-03-10	2023-03-22	2023-04-06	27	15	152864275	152864292	11.0	11.0	0.514	-0.495	1.115	1.365	0.400	1.400	75.00
15	2023-03-23	2023-04-06	2023-04-21	29	15	153414576	152728234	10.5	11.0	0.524	-0.526	0.945	1.525	0.130	2.190	148.00
16	2023-04-10	2023-04-27	2023-05-12	32	15	153592307	153592324	9.0	9.0	0.472	-0.531	0.570	0.875	0.155	1.210	75.00
17	2023-04-28	2023-05-11	2023-05-26	28	15	153705032	153705048	8.0	7.5	0.446	-0.435	0.470	0.415	0.235	0.385	20.50
18	2023-05-12	2023-05-25	2023-06-09	28	15	154169881	154169897	8.0	7.5	0.448	-0.450	0.520	0.565	0.340	0.505	12.00
19	2023-05-26	2023-06-08	2023-06-23	28	15	154277856	154277873	7.5	7.5	0.451	-0.548	0.475	0.825	0.100	1.990	154.00
20	2023-06-09	2023-06-22	2023-07-07	28	15	154750611	154750628	6.0	6.0	0.434	-0.580	0.320	0.570	0.105	0.915	56.00
21	2023-06-26	2023-07-06	2023-07-21	25	15	155372246	155372257	27.0	27.0	0.494	-0.509	1.660	2.075	1.440	1.975	12.00
22	2023-07-07	2023-07-20	2023-08-04	28	15	155436479	155436500	26.5	26.5	0.487	-0.507	1.685	2.315	0.670	2.975	167.50
23	2023-07-21	2023-08-03	2023-08-18	28	15	155372265	155372276	24.0	24.0	0.507	-0.489	1.580	1.900	1.885	0.935	-127.00
24	2023-08-04	2023-08-17	2023-09-01	28	15	155635942	155635963	26.5	26.5	0.513	-0.491	2.205	2.705	1.715	1.975	-24.00
25	2023-08-18	2023-08-31	2023-09-15	28	15	155372289	155372300	26.0	26.0	0.523	-0.473	2.035	2.285	0.195	3.725	328.00
26	2023-09-01	2023-09-14	2023-09-29	28	15	156225663	156225684	21.5	21.5	0.496	-0.501	1.170	1.445	0.405	2.010	133.00
27	2023-09-15	2023-09-28	2023-10-13	28	15	156680248	156680269	21.0	21.0	0.500	-0.498	1.280	1.585	2.345	0.425	-222.50
28	2023-09-29	2023-10-12	2023-10-27	28	15	156777981	156778002	24.0	24.0	0.484	-0.513	1.695	2.395	0.885	2.245	66.00
29	2023-10-13	2023-10-26	2023-11-10	28	15	157309656	157309677	26.5	26.5	0.497	-0.498	2.480	3.650	1.780	2.135	-81.50
30	2023-10-27	2023-11-09	2023-11-24	28	15	157413622	157413643	28.0	28.0	0.499	-0.505	2.345	3.140	0.165	7.450	649.00
31	2023-11-10	2023-11-24	2023-12-08	28	14	157890644	157890657	20.0	20.0	0.519	-0.479	1.250	1.360	0.235	2.820	247.50
32	2023-11-27	2023-12-14	2023-12-29	32	15	158083249	158083270	17.5	17.5	0.509	-0.512	1.435	1.350	0.275	2.290	210.00
33	2023-12-15	2023-12-28	2024-01-12	28	15	158510669	158510690	16.0	16.0	0.524	-0.481	1.040	0.915	0.385	1.085	82.50
34	2023-12-29	2024-01-11	2024-01-26	28	15	158629907	158629928	15.5	15.5	0.510	-0.532	1.200	0.910	0.310	1.135	111.50
35	2024-01-12	2024-01-25	2024-02-09	28	15	159135510	159135531	15.0	15.0	0.499	-0.500	0.890	1.045	0.475	0.890	26.00
36	2024-01-26	2024-02-08	2024-02-23	28	15	159236098	159236119	14.5	14.5	0.529	-0.477	0.895	0.830	0.360	0.985	69.00
37	2024-02-09	2024-02-22	2024-03-08	28	15	159703612	159703633	14.0	14.0	0.519	-0.506	1.015	0.700	0.740	0.400	-2.50
38	2024-02-23	2024-03-07	2024-03-22	28	15	159806471	159806492	14.0	14.0	0.506	-0.491	0.855	1.010	0.670	0.625	-20.00
39	2024-03-08	2024-03-21	2024-04-05	28	15	160272821	160272842	14.5	14.5	0.482	-0.503	0.995	1.565	0.255	1.565	74.00
40	2024-03-22	2024-04-04	2024-04-19	28	15	160140620	160140641	13.0	13.0	0.530	-0.472	0.820	0.680	1.375	0.425	-81.00
41	2024-04-05	2024-04-18	2024-05-03	28	15	160871829	160871850	14.5	14.5	0.496	-0.501	1.050	1.375	1.275	0.625	-97.50
42	2024-04-19	2024-05-02	2024-05-17	28	15	160740051	160740072	16.0	16.0	0.487	-0.514	1.250	1.850	0.150	2.925	217.50
43	2024-05-03	2024-05-16	2024-05-31	28	15	161198907	161198928	12.5	12.5	0.544	-0.459	0.825	0.600	0.275	1.700	165.00
44	2024-05-17	2024-05-30	2024-06-14	28	15	161632355	161632376	11.5	11.5	0.528	-0.495	1.000	0.825	0.675	0.500	0.00
45	2024-05-31	2024-06-13	2024-06-28	28	15	161793013	161793033	12.5	12.0	0.486	-0.516	1.325	1.500	0.175	0.875	52.50
46	2024-06-14	2024-06-27	2024-07-12	28	15	162222887	162222909	11.5	12.0	0.455	-0.560	0.500	1.600	0.250	1.850	50.00
47	2024-06-28	2024-07-11	2024-07-26	28	15	162329219	162329240	11.0	11.0	0.475	-0.509	0.525	0.825	0.225	0.950	42.50
48	2024-07-12	2024-07-25	2024-08-09	28	15	162766361	162766382	10.0	10.0	0.551	-0.446	0.650	0.775	2.850	1.050	-192.50
49	2024-07-26	2024-08-08	2024-08-23	28	15	162848008	162848028	12.0	11.5	0.491	-0.435	0.875	0.950	3.975	0.600	-345.00
50	2024-08-09	2024-08-15	2024-08-30	21	15	162970121	162970142	14.0	14.0	0.504	-0.512	1.375	1.600	0.275	2.825	232.50
51	2024-08-16	2024-09-05	2024-09-20	35	15	159554926	159554948	12.0	12.0	0.444	-0.542	0.875	1.775	1.450	0.525	-182.50
52	2024-09-06	2024-10-03	2024-10-18	42	15	163777795	163777796	15.0	15.0	0.503	-0.501	1.975	3.050	0.600	2.275	60.00
53	2024-10-04	2024-10-31	2024-11-15	42	15	164447577	164447598	14.0	14.0	0.472	-0.507	1.325	2.450	1.200	1.250	-107.50
54	2024-11-01	2024-11-04	2024-11-15	14	11	164447577	164447598	14.0	14.0	0.508	-0.510	1.075	1.100	0.825	1.325	47.50
55	2024-11-05	2024-12-05	2024-12-20	45	15	161398573	161398594	14.0	14.0	0.469	-0.535	1.325	2.525	0.600	3.200	140.00
56	2024-12-06	2025-01-02	2025-01-17	42	15	165457376	16545									

B Short Call on VIXY Options Strategy Trade Log

– PERFORMANCE SUMMARY –

Note: All metrics below are calculated based on a position size of exactly one contract.

Total PnL (\$): 2809.50

Number of completed trades: 67

Average trade PnL (\$): 37.53

Win rate: 76.12%

Trade PnL Standard Deviation (\$): 191.48

Table 2: Short Call on VIXY Options Trade Log (Aug 2022 - Aug 2025)

ID	Entry	Exit	Expiry	DTE (In)	DTE (Out)	Option ID	Strike	Δ (In)	PnL (\$)
1	2022-08-29	2022-09-15	2022-09-30	32	15	148915193	15.0	0.498	47.50
2	2022-09-15	2022-09-29	2022-10-14	29	15	149412453	15.0	0.516	-90.00
3	2022-09-29	2022-10-13	2022-10-28	29	15	149502993	17.5	0.486	67.50
4	2022-10-13	2022-10-27	2022-11-11	29	15	150018470	17.5	0.497	110.00
5	2022-10-27	2022-11-10	2022-11-25	29	15	150107798	15.0	0.515	65.00
6	2022-11-10	2022-11-25	2022-12-09	29	14	150535893	14.0	0.500	57.50
7	2022-11-25	2022-12-08	2022-12-23	28	15	150639375	12.5	0.491	20.00
8	2022-12-08	2022-12-22	2023-01-06	29	15	151077589	12.5	0.471	42.50
9	2022-12-22	2023-01-05	2023-01-20	29	15	143931625	12.0	0.523	67.50
10	2023-01-05	2023-01-19	2023-02-03	29	15	151652599	11.0	0.525	45.50
11	2023-01-19	2023-02-02	2023-02-17	29	15	151497534	10.0	0.568	45.00
12	2023-02-02	2023-02-16	2023-03-03	29	15	151955067	9.0	0.545	7.50
13	2023-02-16	2023-03-02	2023-03-17	29	15	148411794	10.0	0.434	33.50
14	2023-03-02	2023-03-16	2023-03-31	29	15	152523511	9.5	0.453	-58.00
15	2023-03-16	2023-03-30	2023-04-14	29	15	152969722	10.5	0.523	86.00
16	2023-03-30	2023-04-13	2023-04-28	29	15	153083610	9.0	0.534	81.50
17	2023-04-13	2023-04-27	2023-05-12	29	15	153592306	8.5	0.467	26.00
18	2023-04-27	2023-05-11	2023-05-26	29	15	153705032	8.0	0.519	34.00
19	2023-05-11	2023-05-25	2023-06-09	29	15	154169880	7.5	0.553	15.00
20	2023-05-25	2023-06-08	2023-06-23	29	15	154277856	7.5	0.532	54.00
21	2023-06-08	2023-06-22	2023-07-07	29	15	154750611	6.0	0.445	27.50
22	2023-06-22	2023-07-06	2023-07-21	29	15	155274496	10.5	0.441	0.00
23	2023-07-06	2023-07-20	2023-08-04	29	15	155436480	27.0	0.507	149.00
24	2023-07-20	2023-08-03	2023-08-18	29	15	155372265	24.0	0.538	-8.50
25	2023-08-03	2023-08-17	2023-09-01	29	15	155635940	25.5	0.504	-9.00
26	2023-08-17	2023-08-31	2023-09-15	29	15	155372290	27.0	0.503	209.00
27	2023-08-31	2023-09-14	2023-09-29	29	15	156225664	22.0	0.501	113.00
28	2023-09-14	2023-09-28	2023-10-13	29	15	156680246	20.0	0.524	-294.00
29	2023-09-28	2023-10-12	2023-10-27	29	15	156777980	23.5	0.500	76.00
30	2023-10-12	2023-10-26	2023-11-10	29	15	157309649	23.0	0.512	-178.50
31	2023-10-26	2023-11-09	2023-11-24	29	15	157413620	27.0	0.499	206.50
32	2023-11-09	2023-11-24	2023-12-08	29	14	157890645	21.0	0.522	128.50
33	2023-11-24	2023-12-07	2023-12-22	28	15	157966502	18.0	0.457	47.00
34	2023-12-07	2023-12-21	2024-01-05	29	15	158390416	17.5	0.481	34.50
35	2023-12-21	2024-01-04	2024-01-19	29	15	155381402	17.0	0.517	79.00
36	2024-01-04	2024-01-18	2024-02-02	29	15	159032184	16.5	0.475	58.50
37	2024-01-18	2024-02-01	2024-02-16	29	15	158855280	16.0	0.447	33.00
38	2024-02-01	2024-02-15	2024-03-01	29	15	159353593	15.5	0.468	52.50
39	2024-02-15	2024-02-29	2024-03-15	29	15	155381428	15.0	0.437	47.50
40	2024-02-29	2024-03-13	2024-03-28	28	15	159919546	13.5	0.529	16.00
41	2024-03-13	2024-03-28	2024-04-12	30	15	160397390	14.0	0.470	64.00
42	2024-03-28	2024-04-11	2024-04-26	29	15	160528029	13.5	0.472	25.50
43	2024-04-11	2024-04-25	2024-05-10	29	15	160988640	14.0	0.468	17.50
44	2024-04-25	2024-05-09	2024-05-24	29	15	161090539	14.0	0.511	80.00
45	2024-05-09	2024-05-23	2024-06-07	29	15	161529843	12.5	0.468	-50.00
46	2024-05-23	2024-06-06	2024-06-21	29	15	162042432	11.5	0.537	25.00
47	2024-06-06	2024-06-20	2024-07-05	29	15	162123773	11.0	0.574	-12.50
48	2024-06-20	2024-07-05	2024-07-19	29	14	161987044	12.0	0.411	45.00
49	2024-07-05	2024-07-18	2024-08-02	28	15	162447935	10.5	0.502	-57.50
50	2024-07-18	2024-08-01	2024-08-16	29	15	162633975	11.0	0.499	-97.50
51	2024-08-01	2024-08-15	2024-08-30	29	15	162970119	13.0	0.525	117.50
52	2024-08-15	2024-09-05	2024-09-20	36	15	159554926	12.0	0.439	-60.00
53	2024-09-05	2024-10-03	2024-10-18	43	15	163777974	14.0	0.495	72.50
54	2024-10-03	2024-10-31	2024-11-15	43	15	164447577	14.0	0.544	97.50
55	2024-11-01	2024-11-04	2024-11-15	14	11	164447577	14.0	0.508	25.00
56	2024-11-04	2024-12-05	2024-12-20	46	15	161398573	14.0	0.525	100.00
57	2024-12-05	2025-01-02	2025-01-17	43	15	165457377	43.0	0.490	-150.00
58	2025-01-03	2025-01-06	2025-01-17	14	11	165457378	44.0	0.490	17.50
59	2025-01-06	2025-02-06	2025-02-21	46	15	165457401	45.0	0.499	222.50

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Table 2 – continued from previous page

ID	Entry	Exit	Expiry	DTE (In)	DTE (Out)	Option ID	Strike	Δ (In)	PnL (\$)
60	2025-02-06	2025-03-06	2025-03-21	43	15	165457421	43.0	0.513	-600.00
61	2025-03-06	2025-04-02	2025-04-17	42	15	166926157	58.0	0.497	610.00
62	2025-04-02	2025-05-01	2025-05-16	44	15	166926232	51.0	0.489	-855.00
63	2025-05-02	2025-05-05	2025-05-16	14	11	166926241	60.0	0.511	-77.50
64	2025-05-05	2025-06-05	2025-06-20	46	15	166301115	64.0	0.500	482.50
65	2025-06-05	2025-07-03	2025-07-18	43	15	169435018	54.0	0.497	417.50
66	2025-07-03	2025-07-31	2025-08-15	43	15	170101143	48.0	0.494	332.50
67	2025-08-01	2025-08-04	2025-08-15	14	11	170101140	45.0	0.520	177.50

C Iron Condor on VIXY Options Strategy Trade Log

– PERFORMANCE SUMMARY –

Note: All metrics below are calculated based on a position size of exactly one contract.

Total PnL (\$): 748.00

Number of completed trades: 67

Average trade PnL (\$): 11.80

Win rate: 56.72%

Trade PnL Standard Deviation (\$): 59.40

Table 3: Iron Condor on VIXY Options Trade Log (Aug 2022 - Aug 2025)

ID	Entry	Exit	Expiry	DTE (In)	DTE (Out)	Short Call K	Long Call K	Short Put K	Long Put K	PnL (\$)
1	2022-08-29	2022-09-15	2022-09-30	32	15	19.0	25.0	11.0	8.5	12.50
2	2022-09-15	2022-09-29	2022-10-14	29	15	19.0	22.0	13.0	10.0	-37.50
3	2022-09-29	2022-10-13	2022-10-28	29	15	23.0	25.0	14.5	10.5	-2.50
4	2022-10-13	2022-10-27	2022-11-11	29	15	23.0	25.0	15.0	11.0	-40.00
5	2022-10-27	2022-11-10	2022-11-25	29	15	19.5	22.0	13.5	10.0	-22.50
6	2022-11-10	2022-11-25	2022-12-09	29	14	17.0	22.0	12.5	10.5	-12.50
7	2022-11-25	2022-12-08	2022-12-23	28	15	15.5	25.0	11.0	8.0	-7.50
8	2022-12-08	2022-12-22	2023-01-06	29	15	15.5	21.0	11.0	10.0	75.00
9	2022-12-22	2023-01-05	2023-01-20	29	15	15.0	40.0	10.5	7.0	-7.50
10	2023-01-05	2023-01-19	2023-02-03	29	15	15.5	17.5	9.0	7.0	48.00
11	2023-01-19	2023-02-02	2023-02-17	29	15	13.0	18.0	9.0	8.0	-3.00
12	2023-02-02	2023-02-16	2023-03-03	29	15	12.5	14.0	6.5	5.5	-38.50
13	2023-02-16	2023-03-02	2023-03-17	29	15	28.0	30.0	7.0	6.0	-8.00
14	2023-03-02	2023-03-16	2023-03-31	29	15	12.0	14.0	6.5	6.0	2.00
15	2023-03-16	2023-03-30	2023-04-14	29	15	17.0	18.0	8.5	7.5	91.50
16	2023-03-30	2023-04-13	2023-04-28	29	15	13.0	17.0	7.0	6.5	105.50
17	2023-04-13	2023-04-27	2023-05-12	29	15	10.0	12.5	6.5	5.5	0.00
18	2023-04-27	2023-05-11	2023-05-26	29	15	10.0	13.5	7.0	6.5	84.00
19	2023-05-11	2023-05-25	2023-06-09	29	15	10.0	12.5	5.5	5.0	25.50
20	2023-05-25	2023-06-08	2023-06-23	29	15	10.0	13.0	6.5	6.0	-69.00
21	2023-06-08	2023-06-22	2023-07-07	29	15	7.5	10.0	4.0	2.5	-92.50
22	2023-06-22	2023-07-06	2023-07-21	29	15	6.5	12.0	4.0	3.5	0.00
23	2023-07-06	2023-07-20	2023-08-04	29	15	34.0	37.0	23.5	19.0	-46.50
24	2023-07-20	2023-08-03	2023-08-18	29	15	31.0	40.0	21.0	19.0	110.50
25	2023-08-03	2023-08-17	2023-09-01	29	15	37.0	38.0	22.0	19.0	110.50
26	2023-08-17	2023-08-31	2023-09-15	29	15	37.0	40.0	23.0	19.0	-134.50
27	2023-08-31	2023-09-14	2023-09-29	29	15	27.5	33.0	17.0	16.0	107.50
28	2023-09-14	2023-09-28	2023-10-13	29	15	24.5	29.0	18.0	17.5	-89.00
29	2023-09-28	2023-10-12	2023-10-27	29	15	31.0	40.0	20.0	19.0	17.00
30	2023-10-12	2023-10-26	2023-11-10	29	15	29.5	36.0	20.0	19.0	41.00
31	2023-10-26	2023-11-09	2023-11-24	29	15	36.0	40.0	23.0	21.5	-64.50
32	2023-11-09	2023-11-24	2023-12-08	29	14	26.5	32.0	18.5	18.0	6.00
33	2023-11-24	2023-12-07	2023-12-22	28	15	22.0	26.0	14.0	13.0	51.50
34	2023-12-07	2023-12-21	2024-01-05	29	15	21.5	25.0	15.5	15.0	11.50
35	2023-12-21	2024-01-04	2024-01-19	29	15	21.5	26.0	15.0	13.0	112.50
36	2024-01-04	2024-01-18	2024-02-02	29	15	30.0	30.0	14.5	13.5	-2.00
37	2024-01-18	2024-02-01	2024-02-16	29	15	19.0	23.0	14.0	13.0	16.00
38	2024-02-01	2024-02-15	2024-03-01	29	15	18.5	26.0	12.0	9.5	76.00
39	2024-02-15	2024-02-29	2024-03-15	29	15	18.0	23.0	13.0	12.0	49.50
40	2024-02-29	2024-03-13	2024-03-28	28	15	24.0	26.0	12.5	12.0	65.00
41	2024-03-13	2024-03-28	2024-04-12	30	15	19.5	25.0	11.5	11.0	66.00
42	2024-03-28	2024-04-11	2024-04-26	29	15	18.0	20.0	11.5	10.5	-29.00
43	2024-04-11	2024-04-25	2024-05-10	29	15	17.5	21.0	11.0	8.5	25.00
44	2024-04-25	2024-05-09	2024-05-24	29	15	18.0	21.0	12.5	12.0	27.50
45	2024-05-09	2024-05-23	2024-06-07	29	15	14.5	21.0	11.0	7.0	52.50
46	2024-05-23	2024-06-06	2024-06-21	29	15	14.5	23.0	9.0	7.0	25.00
47	2024-06-06	2024-06-20	2024-07-05	29	15	14.0	16.0	9.0	6.5	20.00
48	2024-06-20	2024-07-05	2024-07-19	29	14	14.0	18.0	9.0	8.0	47.50
49	2024-07-05	2024-07-18	2024-08-02	28	15	15.5	19.0	7.5	6.5	25.00
50	2024-07-18	2024-08-01	2024-08-16	29	15	16.0	20.0	10.0	9.0	-10.00
51	2024-08-01	2024-08-15	2024-08-30	29	15	20.0	20.0	10.0	5.5	15.00
52	2024-08-15	2024-09-05	2024-09-20	36	15	16.0	22.0	10.0	5.0	-95.00
53	2024-09-05	2024-10-03	2024-10-18	43	15	19.0	20.0	9.0	8.0	92.50
54	2024-10-03	2024-10-31	2024-11-15	43	15	20.0	25.0	10.0	9.0	20.00
55	2024-11-01	2024-11-04	2024-11-15	14	11	18.0	22.0	12.0	11.0	95.00
56	2024-11-04	2024-12-05	2024-12-20	46	15	26.0	30.0	11.0	10.0	-37.50
57	2024-12-05	2025-01-02	2025-01-17	43	15	60.0	65.0	36.0	35.0	-15.00

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Table 3 – continued from previous page

ID	Entry	Exit	Expiry	DTE (In)	DTE (Out)	Short Call K	Long Call K	Short Put K	Long Put K	PnL (\$)
58	2025-01-03	2025-01-06	2025-01-17	14	11	49.0	60.0	41.0	36.0	-30.00
59	2025-01-06	2025-02-06	2025-02-21	46	15	70.0	75.0	36.0	34.0	92.50
60	2025-02-06	2025-03-06	2025-03-21	43	15	62.0	75.0	36.0	35.0	-102.50
61	2025-03-06	2025-04-02	2025-04-17	42	15	75.0	75.0	42.0	34.0	122.50
62	2025-04-02	2025-05-01	2025-05-16	44	15	70.0	80.0	42.0	39.0	22.50
63	2025-05-02	2025-05-05	2025-05-16	14	11	71.0	78.0	56.0	53.0	-12.50
64	2025-05-05	2025-06-05	2025-06-20	46	15	90.0	110.0	52.0	47.0	-77.50
65	2025-06-05	2025-07-03	2025-07-18	43	15	73.0	75.0	45.0	41.0	-17.50
66	2025-07-03	2025-07-31	2025-08-15	43	15	70.0	75.0	41.0	39.0	-75.00
67	2025-08-01	2025-08-04	2025-08-15	14	11	-	-	-	-	-

D Short Put + Short Call on VIXY Strategy Trade Log

– PERFORMANCE SUMMARY –

Note: All metrics below are calculated based on a position size of exactly one contract.

Total PnL (\$): 2059.00

Number of completed trades: 154

Average trade PnL (\$): 13.32

Win rate: 37.66%

Trade PnL Standard Deviation (\$): 85.03

Table 4: Short Put + Short Call on VIXY Trade Log (Aug 2022 - Aug 2025)

ID	Entry	Exit	Expiry	DTE (In)	DTE (Out)	Strike	Call ID	Put ID	Call Δ	Put Δ	PnL (\$)
1	2022-08-29	2022-09-20	2022-09-30	32	10	15.0	148915193	148915214	0.498	-0.506	137.50
2	2022-09-20	2022-10-06	2022-10-21	31	15	15.0	148521056	148521082	0.520	-0.480	72.50
3	2022-10-06	2022-10-25	2022-11-04	29	10	16.5	149896738	149896759	0.510	-0.485	130.00
4	2022-10-25	2022-11-10	2022-11-25	31	15	16.0	150107800	150107821	0.465	-0.538	-22.50
5	2022-11-10	2022-11-29	2022-12-09	29	10	14.0	150535893	150535914	0.500	-0.496	-2.50
6	2022-11-29	2022-12-14	2022-12-30	31	16	12.5	150746248	150746269	0.538	-0.460	7.50
7	2022-12-15	2022-12-16	2023-01-13	29	28	12.0	151179419	151179440	0.558	-0.444	0.00
8	2022-12-16	2022-12-19	2023-01-13	28	25	13.0	151179421	151179442	0.364	-0.631	0.00
9	2022-12-19	2023-01-10	2023-01-20	32	10	12.0	143931625	143931641	0.512	-0.490	-8.50
10	2023-01-13	2023-01-17	2023-02-10	28	24	9.5	151754041	151754062	0.592	-0.411	0.00
11	2023-01-20	2023-01-23	2023-02-17	28	25	11.0	151497535	151497556	0.352	-0.663	0.00
12	2023-01-31	2023-02-01	2023-03-17	45	44	10.0	148411794	148411815	0.428	-0.575	0.00
13	2023-02-01	2023-02-03	2023-02-24	23	21	9.0	151837575	151837596	0.490	-0.514	5.00
14	2023-02-07	2023-02-08	2023-03-03	24	23	9.0	151955067	151955088	0.522	-0.488	0.00
15	2023-02-10	2023-02-13	2023-03-17	35	32	10.0	148411794	148411815	0.505	-0.481	0.00
16	2023-02-14	2023-02-16	2023-03-17	31	29	12.0	148411796	148411817	0.177	-0.802	-3.00
17	2023-02-17	2023-02-23	2023-03-17	28	22	11.0	148411795	148411816	0.296	-0.733	24.00
18	2023-03-06	2023-03-07	2023-04-06	31	30	10.0	152864273	152864290	0.276	-0.770	0.00
19	2023-03-15	2023-03-16	2023-04-14	30	29	11.5	152969724	152969741	0.515	-0.483	0.00
20	2023-03-16	2023-03-20	2023-04-14	29	25	11.0	152969723	152969740	0.468	-0.529	15.00
21	2023-03-20	2023-03-22	2023-04-21	32	30	10.5	153414576	153414588	0.579	-0.416	-16.50
22	2023-03-24	2023-03-28	2023-04-21	28	24	11.0	152728216	152728234	0.426	-0.588	13.50
23	2023-03-29	2023-03-30	2023-04-28	30	29	9.0	153083610	153083627	0.549	-0.454	0.00
24	2023-03-31	2023-04-11	2023-04-28	28	17	9.0	153083610	153083627	0.539	-0.455	40.00
25	2023-04-11	2023-04-13	2023-05-12	31	29	9.0	153592307	153592324	0.450	-0.546	4.50
26	2023-04-13	2023-06-01	2023-06-16	64	15	9.0	149857280	149857308	0.480	-0.542	-7.50
27	2023-06-02	2023-06-08	2023-06-30	28	22	6.5	154394003	154394020	0.459	-0.540	-14.50
28	2023-06-09	2023-06-12	2023-07-07	28	25	6.0	154750611	154750628	0.434	-0.580	0.00
29	2023-06-12	2023-06-14	2023-07-14	32	30	7.5	154860325	154860342	0.211	-0.819	4.00
30	2023-06-14	2023-06-15	2023-07-14	30	29	7.0	154860324	154860341	0.189	-0.817	0.00
31	2023-06-15	2023-06-20	2023-07-14	29	24	6.5	154860323	154860340	0.313	-0.721	-12.00
32	2023-06-20	2023-06-26	2023-07-21	31	25	5.5	155274499	155274507	0.541	-0.463	1.50
33	2023-06-26	2023-07-11	2023-07-21	25	10	27.0	155372246	155372257	0.494	-0.509	107.00
34	2023-07-11	2023-07-17	2023-08-11	31	25	26.0	155436520	155436541	0.508	-0.492	-10.00
35	2023-07-17	2023-08-08	2023-08-18	32	10	24.0	155372265	155372276	0.536	-0.463	132.00
36	2023-08-08	2023-08-29	2023-09-08	31	10	25.5	155999644	155999665	0.511	-0.491	191.00
37	2023-08-29	2023-08-30	2023-09-29	31	30	23.0	156225666	156225687	0.496	-0.500	0.00
38	2023-09-01	2023-09-19	2023-09-29	28	10	21.5	156225663	156225684	0.496	-0.501	57.50
39	2023-09-19	2023-10-10	2023-10-20	31	10	21.0	156434360	156434382	0.480	-0.521	115.50
40	2023-10-10	2023-10-27	2023-11-10	31	14	23.0	157309649	157309670	0.501	-0.498	-26.00
41	2023-10-27	2023-11-09	2023-11-24	28	15	28.0	157413622	157413643	0.499	-0.505	-280.00
42	2023-11-09	2023-11-17	2023-12-08	29	21	21.0	157890645	157890658	0.522	-0.483	-25.50
43	2023-11-17	2023-12-05	2023-12-15	28	10	19.0	155381376	155381390	0.485	-0.520	31.00
44	2023-12-05	2023-12-06	2024-01-05	31	30	17.5	158390416	158390437	0.479	-0.549	0.00
45	2023-12-06	2023-12-07	2024-01-05	30	29	19.0	158390419	158390440	0.355	-0.711	0.00
46	2023-12-07	2023-12-12	2024-01-05	29	24	16.5	158390414	158390435	0.629	-0.369	29.50
47	2023-12-12	2023-12-15	2024-01-12	31	28	16.5	158510670	158510691	0.467	-0.550	-0.50
48	2023-12-15	2023-12-18	2024-01-12	28	25	16.0	158510669	158510690	0.524	-0.481	0.00
49	2023-12-18	2023-12-20	2024-01-19	32	30	16.5	158949554	158949561	0.501	-0.514	8.50
50	2023-12-20	2024-01-08	2024-01-19	30	11	18.0	155381403	155381417	0.482	-0.545	-11.00
51	2024-01-08	2024-01-09	2024-02-09	32	31	15.0	159135510	159135531	0.548	-0.453	0.00
52	2024-01-09	2024-01-19	2024-02-09	31	21	15.5	159135511	159135532	0.437	-0.563	46.50
53	2024-01-19	2024-02-06	2024-02-16	28	10	16.0	158855280	158855301	0.382	-0.613	-1.00
54	2024-02-06	2024-02-09	2024-03-08	31	28	14.0	159703612	159703633	0.512	-0.491	-15.00
55	2024-02-09	2024-02-12	2024-03-08	28	25	14.5	159703613	159703634	0.430	-0.561	0.00
56	2024-02-12	2024-02-13	2024-03-15	32	31	15.0	155381428	155381442	0.434	-0.551	0.00
57	2024-02-13	2024-02-14	2024-03-15	31	30	18.0	155381431	155381445	0.320	-0.785	0.00

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Table 4 – continued from previous page

ID	Entry	Exit	Expiry	DTE (In)	DTE (Out)	Strike	Call ID	Put ID	Call Δ	Put Δ	PnL (\$)
58	2024-02-14	2024-02-29	2024-03-15	30	15	15.0	155381428	155381442	0.464	-0.547	32.00
59	2024-02-29	2024-03-05	2024-03-28	28	23	13.5	159919546	159919567	0.529	-0.468	13.50
60	2024-03-05	2024-03-06	2024-04-05	31	30	14.5	160272821	160272842	0.476	-0.530	0.00
61	2024-03-07	2024-03-08	2024-04-05	29	28	16.0	160272824	160272845	0.320	-0.651	0.00
62	2024-03-11	2024-03-12	2024-04-12	32	31	15.0	160397392	160397413	0.461	-0.542	0.00
63	2024-03-12	2024-03-14	2024-04-12	31	29	13.0	160397388	160397409	0.592	-0.398	-6.50
64	2024-03-14	2024-03-15	2024-04-12	29	28	16.5	160397395	160397416	0.294	-0.656	0.00
65	2024-03-15	2024-03-19	2024-04-12	28	24	14.0	160397390	160397411	0.524	-0.490	14.00
66	2024-03-19	2024-03-20	2024-04-19	31	30	13.0	160140620	160140641	0.570	-0.430	0.00
67	2024-03-20	2024-03-22	2024-05-17	58	56	13.0	160740048	160740069	0.546	-0.455	6.00
68	2024-03-22	2024-03-25	2024-04-19	28	25	13.5	160810571	160810584	0.448	-0.571	0.00
69	2024-03-25	2024-03-27	2024-04-19	25	23	15.0	160140622	160140643	0.254	-0.813	14.50
70	2024-03-28	2024-04-04	2024-04-19	22	15	13.0	160140620	160140641	0.501	-0.495	8.00
71	2024-04-05	2024-04-23	2024-05-03	28	10	14.5	160871829	160871850	0.496	-0.501	100.00
72	2024-04-23	2024-04-30	2024-05-24	31	24	14.0	161090539	161090560	0.486	-0.519	30.00
73	2024-04-30	2024-05-01	2024-05-31	31	30	13.5	161198909	161198930	0.531	-0.467	0.00
74	2024-05-02	2024-05-03	2024-05-31	29	28	13.5	161198909	161198930	0.459	-0.560	0.00
75	2024-05-03	2024-05-06	2024-05-31	28	25	12.5	161198907	161198928	0.544	-0.459	0.00
76	2024-05-13	2024-05-14	2024-06-14	32	31	11.5	161632355	161632376	0.618	-0.357	0.00
77	2024-05-14	2024-05-15	2024-06-14	31	30	12.5	161632357	161632378	0.400	-0.693	0.00
78	2024-05-15	2024-05-16	2024-06-14	30	29	12.0	161632356	161632377	0.419	-0.596	0.00
79	2024-05-20	2024-05-23	2024-06-21	32	29	12.0	159752905	159752916	0.428	-0.579	-0.00
80	2024-05-23	2024-05-28	2024-06-21	29	24	13.0	159752906	159752917	0.321	-0.679	-10.00
81	2024-06-17	2024-06-18	2024-07-19	32	31	13.0	161987045	161987066	0.266	-0.742	0.00
82	2024-06-20	2024-06-24	2024-07-19	29	25	13.0	161987045	161987066	0.293	-0.699	-12.50
83	2024-06-26	2024-06-28	2024-07-19	23	21	11.0	161987043	161987064	0.457	-0.541	2.50
84	2024-07-01	2024-07-02	2024-08-16	46	45	11.0	162633975	162633996	0.436	-0.563	0.00
85	2024-07-03	2024-07-05	2026-01-16	562	560	40.0	156866797	156866821	0.350	-0.804	0.00
86	2024-07-10	2024-07-11	2024-08-16	37	36	12.0	162633976	162633997	0.257	-0.749	0.00
87	2024-07-11	2024-07-15	2024-08-16	36	32	10.0	162633974	162633995	0.596	-0.404	-5.00
88	2024-07-16	2024-07-17	2024-08-16	31	30	12.0	162633976	162633997	0.238	-0.762	0.00
89	2024-07-17	2024-07-18	2024-08-16	30	29	11.0	162633975	162633996	0.449	-0.566	0.00
90	2024-08-02	2024-08-05	2024-08-30	28	25	15.0	162970123	162970144	0.577	-0.403	0.00
91	2024-08-05	2024-08-06	2024-09-20	46	45	26.0	159554940	159554962	0.495	-0.441	0.00
92	2024-08-12	2024-09-10	2024-09-20	39	10	13.0	159554927	159554949	0.573	-0.430	147.50
93	2024-09-12	2024-09-24	2024-10-18	36	24	13.0	163777973	163777994	0.476	-0.535	87.50
94	2024-09-24	2024-09-25	2024-10-18	24	23	11.0	163777971	163777992	0.652	-0.374	0.00
95	2024-09-25	2024-09-27	2024-10-18	23	21	13.0	163777973	163777994	0.332	-0.654	10.00
96	2024-10-01	2024-10-02	2024-11-15	45	44	14.0	164447577	164447598	0.472	-0.505	0.00
97	2024-10-03	2024-10-04	2024-11-15	43	42	12.0	164447575	164447596	0.704	-0.288	0.00
98	2024-10-09	2024-10-11	2024-11-15	37	35	14.0	164447577	164447598	0.492	-0.515	0.00
99	2024-10-11	2024-10-14	2024-11-15	35	32	13.0	164447576	164447597	0.539	-0.512	0.00
100	2024-10-15	2024-10-31	2024-11-15	31	15	13.0	164447576	164447597	0.537	-0.479	25.00
101	2024-10-31	2024-11-08	2024-12-20	50	42	14.0	161398573	161398594	0.547	-0.453	-45.00
102	2024-11-08	2024-11-12	2025-01-17	70	66	45.0	165457379	165457390	0.534	-0.457	110.00
103	2024-11-12	2024-11-13	2024-12-20	38	37	45.0	165457357	165457368	0.497	-0.498	0.00
104	2024-11-13	2024-11-15	2024-12-20	37	35	44.0	165457356	165457367	0.495	-0.499	62.50
105	2024-11-15	2024-11-19	2024-12-20	35	31	47.0	165457359	165457370	0.499	-0.502	42.50
106	2024-11-19	2024-11-20	2024-12-20	31	30	52.0	165479727	165479737	0.311	-0.654	0.00
107	2024-11-20	2024-11-21	2024-12-20	30	29	48.0	165457360	165457371	0.503	-0.490	0.00
108	2024-11-22	2024-11-25	2024-12-20	28	25	46.0	165457358	165457369	0.482	-0.513	0.00
109	2024-11-25	2024-11-27	2024-12-20	25	23	45.0	165457357	165457368	0.452	-0.578	-10.00
110	2024-11-27	2024-11-29	2024-12-20	23	21	50.0	165479725	165479735	0.245	-0.810	0.00
111	2024-11-29	2024-12-03	2024-12-20	21	17	42.0	165457354	165457365	0.528	-0.472	10.00
112	2024-12-11	2024-12-12	2025-01-17	37	36	50.0	165479745	165479755	0.305	-0.741	0.00
113	2024-12-13	2024-12-18	2025-01-17	35	30	42.0	165457376	165457387	0.531	-0.467	67.50
114	2024-12-19	2024-12-20	2025-01-17	29	28	57.0	166301045	166301055	0.498	-0.484	0.00
115	2024-12-20	2024-12-23	2025-01-17	28	25	49.0	165457383	165457394	0.480	-0.538	0.00
116	2024-12-23	2024-12-31	2025-01-17	25	17	45.0	165457379	165457390	0.483	-0.519	27.50
117	2024-12-31	2025-01-24	2025-02-21	52	28	48.0	165457404	165457415	0.476	-0.534	72.50
118	2025-01-24	2025-01-27	2025-02-21	28	25	42.0	165457398	165457409	0.486	-0.515	0.00
119	2025-01-27	2025-01-31	2025-02-21	25	21	45.0	165457401	165457412	0.489	-0.528	45.00
120	2025-01-31	2025-02-11	2025-02-21	21	10	43.0	165457399	165457410	0.547	-0.450	222.50
121	2025-02-11	2025-02-12	2025-03-21	38	37	44.0	165457422	165457433	0.482	-0.519	0.00
122	2025-02-12	2025-02-21	2025-03-21	37	28	41.0	165457419	165457430	0.571	-0.426	120.00
123	2025-02-21	2025-02-25	2025-03-21	28	24	42.0	165457420	165457431	0.623	-0.377	52.50
124	2025-02-25	2025-02-28	2025-03-21	24	21	45.0	165457423	165457434	0.494	-0.511	-80.00
125	2025-02-28	2025-03-03	2025-03-21	21	18	50.0	165479785	165479795	0.345	-0.722	0.00
126	2025-03-04	2025-03-14	2025-04-17	44	34	53.0	166926152	166926193	0.496	-0.505	300.00
127	2025-03-14	2025-04-04	2025-04-17	34	13	52.0	166926151	166926192	0.488	-0.506	-215.00

Continued on next page

Table 4 – continued from previous page

ID	Entry	Exit	Expiry	DTE (In)	DTE (Out)	Strike	Call ID	Put ID	Call Δ	Put Δ	PnL (\$)
128	2025-04-04	2025-04-09	2025-05-16	42	37	80.0	167818978	167818981	0.535	-0.471	-570.00
129	2025-04-09	2025-04-15	2025-05-16	37	31	70.0	167818976	167818979	0.486	-0.502	275.00
130	2025-04-15	2025-04-21	2025-05-16	31	25	68.0	168585629	168585646	0.537	-0.470	100.00
131	2025-04-21	2025-04-23	2025-05-16	25	23	77.0	168585636	168585653	0.502	-0.504	135.00
132	2025-04-23	2025-05-01	2025-05-16	23	15	70.0	167818976	167818979	0.478	-0.529	40.00
133	2025-05-01	2025-05-13	2025-06-20	50	38	65.0	165730266	165730267	0.497	-0.509	-210.00
134	2025-05-13	2025-05-15	2025-06-20	38	36	52.0	165479807	165479817	0.489	-0.512	-85.00
135	2025-05-15	2025-05-21	2025-06-20	36	30	54.0	165479809	165479819	0.429	-0.552	-2.50
136	2025-05-21	2025-06-03	2025-06-20	30	17	55.0	165479810	165479820	0.516	-0.475	265.00
137	2025-06-03	2025-06-05	2025-07-18	45	43	60.0	169435024	169435045	0.340	-0.777	-127.50
138	2025-06-05	2025-06-06	2025-07-18	43	42	55.0	169435019	169435040	0.461	-0.535	0.00
139	2025-06-06	2025-06-12	2025-07-18	42	36	51.0	169435015	169435036	0.493	-0.497	75.00
140	2025-06-12	2025-06-20	2025-07-18	36	28	52.0	169435016	169435037	0.465	-0.524	110.00
141	2025-06-20	2025-06-24	2025-07-18	28	24	60.0	169435024	169435045	0.336	-0.665	57.50
142	2025-06-24	2025-06-25	2025-07-18	24	23	50.0	169435014	169435035	0.420	-0.595	0.00
143	2025-06-25	2025-06-27	2025-07-18	23	21	52.0	169435016	169435037	0.323	-0.670	57.50
144	2025-06-27	2025-06-30	2025-07-18	21	18	47.0	169435011	169435032	0.502	-0.517	0.00
145	2025-06-30	2025-07-02	2025-08-15	46	44	47.0	170101142	170101167	0.552	-0.458	-80.00
146	2025-07-02	2025-07-03	2025-08-15	44	43	55.0	170101150	170101175	0.321	-0.702	0.00
147	2025-07-03	2025-07-07	2025-08-15	43	39	56.0	170101151	170101176	0.286	-0.731	0.00
148	2025-07-08	2025-07-16	2025-08-15	38	30	46.0	170101141	170101166	0.510	-0.482	27.50
149	2025-07-16	2025-07-18	2025-08-15	30	28	51.0	170101146	170101171	0.349	-0.692	80.00
150	2025-07-18	2025-07-22	2025-08-15	28	24	45.0	170101140	170101165	0.485	-0.524	-27.50
151	2025-07-22	2025-07-23	2025-08-15	24	23	44.0	170101139	170101164	0.504	-0.493	0.00
152	2025-07-23	2025-07-24	2025-08-15	23	22	45.0	170101140	170101165	0.377	-0.635	0.00
153	2025-07-24	2025-08-05	2025-08-15	22	10	42.0	170101137	170101162	0.510	-0.494	110.00
154	2025-08-05	2025-08-28	2025-09-19	45	22	44.0	166849372	166849393	0.484	-0.509	-135.00

E Implied Borrow Cost Calculation (6-Month Spread Sample)

Implied Borrow Cost Data (First 20 Rows)

Date	Strike	A_1	A_2	T_1	T_2	Implied BC (%)
2022-08-29	12.0	14.5278	14.1077	0.0877	0.5479	6.37
2022-08-29	13.0	14.4238	14.1584	0.0877	0.5479	4.04
2022-08-29	14.0	14.5699	14.2090	0.0877	0.5479	5.45
2022-08-29	15.0	14.6159	14.2347	0.0877	0.5479	5.74
2022-08-29	16.0	14.5120	14.2103	0.0877	0.5479	4.56
2022-08-29	17.0	14.4331	14.1859	0.0877	0.5479	3.75
2022-08-29	18.0	14.4541	14.2366	0.0877	0.5479	3.29
2022-08-29	19.0	14.4752	14.1372	0.0877	0.5479	5.13
2022-08-29	20.0	14.3713	14.1879	0.0877	0.5479	2.79
2022-08-29	21.0	14.5423	14.1135	0.0877	0.5479	6.50
2022-08-29	22.0	14.5384	14.1642	0.0877	0.5479	5.67
2022-08-29	23.0	14.3594	14.1398	0.0877	0.5479	3.35
2022-08-29	24.0	14.4305	14.1905	0.0877	0.5479	3.64
2022-08-29	25.0	14.4266	14.1911	0.0877	0.5479	3.58
2022-08-30	12.0	14.8292	14.5342	0.0849	0.5452	4.37
2022-08-30	13.0	14.8254	14.4849	0.0849	0.5452	5.05
2022-08-30	14.0	14.7716	14.5107	0.0849	0.5452	3.87
2022-08-30	15.0	14.8428	14.4865	0.0849	0.5452	5.28
2022-08-30	16.0	14.8390	14.5622	0.0849	0.5452	4.09
2022-08-30	17.0	14.8102	14.5380	0.0849	0.5452	4.03

- A_1 is the synthetic forward price calculated using the options data for the first expiration date (the near-term expiration).
- A_2 is the synthetic forward price calculated using the options data for the second expiration date (the longer-term expiration).
- T_1 is the time until expiration for the near-term options.
- T_2 is the time until expiration for the longer-term options.

F AI-Assisted Research Prompt Log

This appendix records the generative-AI prompts used during the project to scaffold the methodology and accelerate the implementation. Prompts are grouped by project phase and can be cited inline from the main text as P n .

Design and Methodology

Prompt P1 — Framing the research question.

I want to study the ProShares VIXY ETF. It clearly decays over time because the underlying VIX-futures curve is usually in contango and the fund has to roll its front-month into the next month every day. I want a short-VIXY strategy, but I've read that VIXY is classified as hard-to-borrow and the borrow fee often eats the whole edge. Explain (a) why the fund structurally decays even when realized volatility is flat, (b) why institutional traders use a long-put + short-call synthetic short instead of borrowing shares, and (c) why the same supply/demand that drives securities-lending fees is reflected in relative put-vs-call prices. I'll use this framing to write the introduction and motivation sections of my report.

Prompt P2 — Deriving the two-horizon implied borrow cost.

Starting from the no-arbitrage put-call parity $C - P = (F - K)e^{-rT}$, walk me through the full derivation of an implied borrow cost b that only uses option prices — no observed spot S . Assume the forward is $F_i = Se^{(r-b)T_i}$. Take two matched call-put pairs at the same date but with different maturities $T_1 < T_2$, form the discounted-forward identity $C_i - P_i + Ke^{-rT_i} = Se^{-bT_i}$, and show how taking the ratio eliminates S . Conclude with the closed-form expression $b = [\ln(C_1 - P_1 + Ke^{-rT_1}) - \ln(C_2 - P_2 + Ke^{-rT_2})]/(T_2 - T_1)$. Explain in one paragraph why this is robust to ETF/option close-time mismatch.

Prompt P3 — Choosing the horizon pair.

I have OptionMetrics data and want to pick the best (T_1, T_2) pair for the two-horizon borrow-cost estimator. I plan to fix $T_1 = 1$ month and test $T_2 \in \{3, 6, 12\}$ months. Give me a checklist for deciding: cross-sectional dispersion across strikes, day-to-day noise, liquidity near at-the-money, and whether long-dated quotes pick up liquidity or term-premium distortions. Recommend a default and justify it so I can put the choice in the methodology section of my paper.

Data Preparation and Cleaning

Prompt P4 — OptionMetrics schema handling.

I have an OptionMetrics CSV for VIXY options from WRDS with columns `date`, `exdate`, `cp_flag`, `strike_price`, `best_bid`, `best_offer`, `delta`, `contract_size`, `optionid`. Write defensive pandas code that: (1) parses dates, (2) keeps only `contract_size == 100`, (3) divides `strike_price` by 1000 when its max exceeds 5000 (OptionMetrics scaling), (4) drops rows with missing bid/ask/delta or crossed quotes, (5) builds `mid = (bid + ask)/2`, `dte_days = (exdate - date).dt.days`, `T = dte_days/365`, and (6) drops wide quotes with $(ask - bid)/mid > 0.50$. Keep the code short and return one cleaned DataFrame.

Prompt P5 — Matching call-put pairs for a target DTE and delta.

Given the cleaned OptionMetrics DataFrame, write a helper `select_target_expiry(group, target_days, tolerance=20)` that, for each trade date, finds the single expiry whose DTE is closest to `target_days` (within the tolerance). Then, for that expiry, merge calls and puts on `[date, exdate, strike]` to build same-strike pairs. For the synthetic-short backtest I also need a secondary scoring function that picks the single strike minimising $|call_delta - 0.50| + |put_delta + 0.50|$ so both legs are jointly near-ATM. Return both helpers with docstrings.

Backtest Engines

Prompt P6 — Synthetic short, daily-PnL engine.

Build a backtest for a one-contract VIXY synthetic short: long the $\sim -0.50\Delta$ put and short the $\sim +0.50\Delta$ call at the same strike on an expiry close to 30 DTE. On every trading day: (a) mark both legs at mid, (b) carry the last valid quote forward if one leg is temporarily missing, (c) roll the position when `remaining_dte` $\leq$ `ROLL_DTE` (configurable, I'll start at 7–15), (d) charge an optional `COMMISSION_PER_LEG` on entry and exit. Emit two DataFrames: a trade log (entry/exit dates, strikes, deltas, entry/exit mids, realized PnL in dollars) and a daily-PnL series. The daily PnL should sum across legs and use the 100-share multiplier.

Prompt P7 — Reusing the engine for short-call and iron-condor.

Refactor the synthetic-short engine so the leg-selection and position-sign logic is pluggable. Then use it to implement two more strategies: (1) a naked short call at $+0.50\Delta$, 30-DTE entry, roll when DTE drops below 7; (2) an iron condor at the same expiry with short $\pm 0.20\Delta$ legs and long $\pm 0.10\Delta$ protective wings. Keep the entry/exit/roll plumbing identical so the three strategies are directly comparable. For each strategy, output a trade-level PnL table and a daily PnL series keyed on trade date.

Prompt P8 — Trade-level and period-level performance metrics.

Given a trade-log DataFrame with a realized PnL column and a daily-PnL series, compute: (1) total PnL, number of completed trades, average trade PnL, win rate (fraction with PnL > 0), and trade-level standard deviation; (2) annualised Sharpe within each calendar-year window `mean(daily_pnl) / sd(daily_pnl) * sqrt(252)`, computed independently for year 1, year 2, and year 3 of the backtest; (3) a consolidated pandas table across my three strategies (Synthetic Short, Short Call, Iron Condor) that I can paste into LaTeX with `to_latex`.

Borrow-Cost Analysis

Prompt P9 — Single-date Bloomberg-style parity verification.

Before I run the full panel, I want a one-date sanity check. Given inputs $S = 33.53$, $K = 34$, $C_{mid} = 3.70$, $P_{mid} = 4.30$, $T = 45/365$, $r = 3.683\%$ (from a Bloomberg snapshot and a same-day Treasury yield), write a short Python function `calculate_implied_borrow_cost(spot, strike, call_mid, put_mid, days_to_expiry, risk_free_rate)` that returns the implied forward $F = K + (C - P)e^{rT}$ and the implied borrow $b = r - \ln(F/S)/T$. Explain in one paragraph why this is just the single-maturity limit of the more general two-horizon formula.

Prompt P10 — Two-horizon cross-section and time series.

Implement `run_borrow_cost_analysis(options_df, r=0.045, T1_days=30, T2_days=182)` that: (1) cleans `OptionMetrics` data; (2) builds same-date same-strike call–put pairs; (3) computes $A = C - P + Ke^{-rT}$ for each pair; (4) for every trading date, uses `select_target_expiry` to extract two expiry buckets (T_1 and T_2); (5) inner-joins on `[date, strike]` so the same strike appears in both buckets; (6) computes $bc = (\ln A_1 - \ln A_2) / (T_2 - T_1)$; (7) returns one DataFrame with columns `[date, strike, A_1, A_2, T_1, T_2, bc_pct]`. Add an option to output the median-across-strikes per date for the term-structure plot.

Prompt P11 — Comparing horizon choices and diagnosing noise.

Run the same two-horizon estimator with $T_2 \in \{91, 182, 365\}$ and plot, on a common axis, the cross-sectional dispersion across strikes for one representative date. Explain the trade-off I should see: the 3-month pair is noisier and more sensitive to microstructure, the 12-month pair is smoother but embeds extra term-premium effects, and 6-month is the compromise. Give me matplotlib code that produces the two-panel figure the paper uses (cross-section across strikes on the left, median time series on the right) with the same labels and captions.

Results, Visualization, and Writeup

Prompt P12 — Cumulative-PnL and log-return comparison plots.

I have three daily-PnL DataFrames (`daily_synth`, `daily_short_call`, `daily_condor`) and a VIXY price series from Nasdaq (`Close/Last`). Produce two comparison plots. First: cumulative dollar PnL on one axis for all three strategies over Aug 2022–Aug 2025. Second: cumulative log-return for each strategy using a notional anchor of \$2,000 (to convert dollar PnL into a return), overlaid with the inverted VIXY cumulative log return so the short-VIXY exposure reads visually. Use `.to_numpy()` on both arrays to avoid the matplotlib/pandas dtype warnings.

All code snippets and plots produced in response to these prompts were reviewed, adapted, and empirically validated by the authors before being incorporated into the results reported.